

What the Kernel Measures: Bounding Feature Geometry for Few-Shot Molecular Property Prediction

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Abstract

Few-shot molecular property prediction increasingly uses deep-kernel Gaussian processes, which combine a neural feature map with a base kernel and adapt to a new assay from only a small support set. In this setting, the geometry presented to the kernel remains unconstrained, creating two failure modes. An unbounded feature map permits large rescalings that confound the learned geometry with the fitted length-scale. Direct concatenation of molecular graph embeddings and substructure fingerprints produces an incommensurable representation in which the high-dimensional fingerprint dominates the distance metric. We propose the Bounded Output Neural Dendrite (BOND), a branch-structured feature map inspired by dendritic integration. Each output coordinate aggregates a fixed number of bounded responses through range-limited weights, imposing a structural bound without post hoc normalisation. We prove that BOND bounds the diameter of the feature space by a data-independent constant, thereby excluding the dispersion regime in which the kernel length-scale becomes unidentifiable. Empirically, fingerprints account for between 83% and 92% of the squared pairwise distance at the input to the feature map. BOND leaves the encoder, Gaussian process, and meta-learning procedure unchanged. Across multiple random seeds on the MoleculeNet few-shot benchmark, it outperforms state-of-the-art baselines on Tox21 and SIDER under both train-from-scratch and pre-trained protocols. Direct geometric measurements recover both predicted degeneracy regimes but also qualify our explanation. Median-based length-scale initialisation absorbs most scale variation, indicating that performance is largely robust to the feature map rather than driven by it. Code and further details are available at <https://zhangzhstu.github.io/bond/>.

CCS Concepts

• **Computing methodologies** → *Gaussian processes*; • **Applied computing** → *Bioinformatics*.

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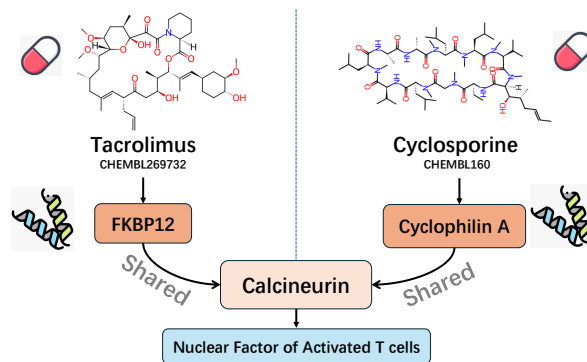


Figure 1: Substructure overlap is not a reliable proxy for behaviour. Tacrolimus and cyclosporine are structurally dissimilar yet inhibit the same pathway, so a distance built on substructure overlap places them far apart although they act alike.

Keywords

Few-shot learning, Molecular property prediction, Deep kernel learning, Gaussian processes, Meta-learning, Drug discovery

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1 Introduction

Screening candidate molecules remains the rate-limiting step in early-stage drug discovery [11, 19, 24]. Candidates are inexpensive to enumerate but costly to assay, so a newly proposed target typically comes with only a handful of measured actives rather than the tens of thousands of labels a conventional deep model

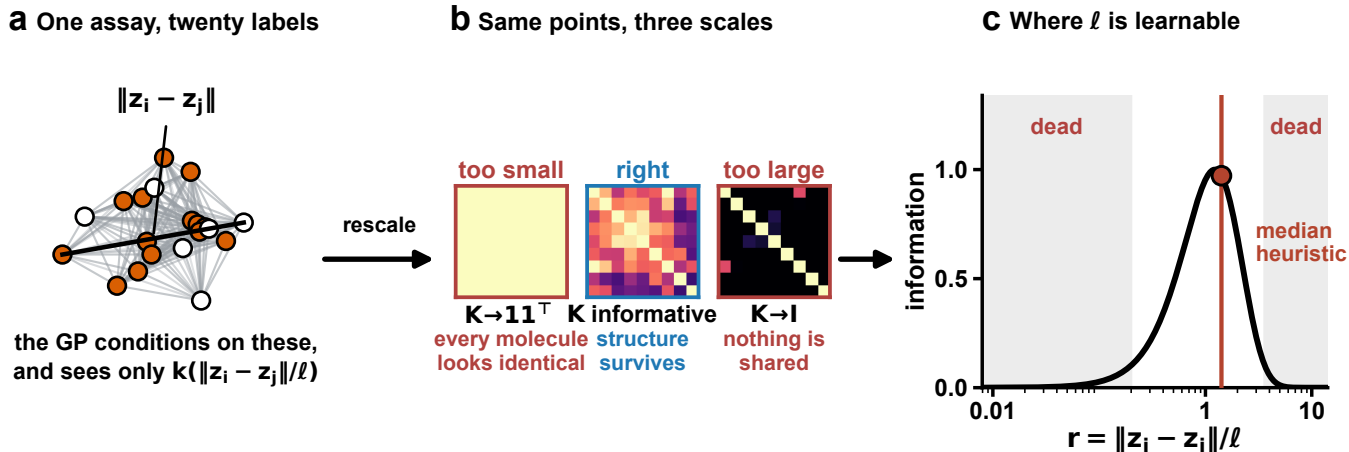


Figure 2: Why the geometry of the kernel’s input space is the object to manage. (a) A Gaussian process reaches the support set only through the scaled pairwise distance $r_{ij} = \|z_i - z_j\|/\ell$. (b) Rescaling alone therefore changes the kernel completely: the same nine molecules at three scales give a rank-one K , an informative K , and the identity. The matrices are computed, not drawn. (c) The length-scale carries Fisher information proportional to $g(r)^2$ with $g(r) = -r k'(r)$, which vanishes at both ends; shaded regions are where it stops being identifiable.

requires [16, 31, 32]. This imbalance makes few-shot learning a natural formulation: a model learns across many historical assays and then adapts to a new one from a support set of perhaps twenty molecules.

Within this setting, deep-kernel Gaussian processes have emerged as a leading class of methods [6]. A Gaussian process specialises to a new assay without gradient-based adaptation, because conditioning on the support set is a closed-form operation that provides calibrated uncertainty and only a small number of kernel hyper-parameters are fitted per assay. The neural network instead learns the representation in which the kernel measures distance, and shares it across assays.

This division of labour places considerable importance on an object that has received little attention in the literature: *the geometry of the representation the kernel sees*. Every conclusion a Gaussian process draws about a new assay is mediated by the pairwise distances $\|z_i - z_j\|$ between the mapped support molecules, and the features and the length-scale enter the kernel only through their ratio. Figure 2 illustrates the consequence. Holding the molecules fixed and changing only the scale of the map, the same nine points can produce a kernel that is rank one, informative, or the identity, and the length-scale is estimable only at the intermediate scale; panel (c) states this in closed form, and Section 4 formalises it. Prior work has focused on the encoder, the meta-learning loop, and the relationships among assays, leaving the geometry of the space in which these distances are measured to the optimiser. In the few-shot regime this omission is consequential, and it gives rise to two specific failure modes.

The feature map is unbounded. With only twenty support points, the inner problem fits the kernel hyper-parameters while the outer problem simultaneously reshapes the features. Because the marginal likelihood depends on the features and the length-scale only through the ratio $\|z_i - z_j\|/\ell$, it cannot distinguish a genuinely

improved representation from one that has merely been rescaled, and a linear or ReLU head imposes no constraint that keeps such rescaling bounded. As the scale diverges, the length-scale ceases to be identifiable, the curvature of the inner objective in that direction approaches zero, and the implicit-function-theorem hypergradient required by the outer loop becomes ill-posed.

Two representations are concatenated that are not commensurable. Molecular graphs and substructure fingerprints provide complementary information, and Figure 1 shows a case in which they disagree. The standard way to combine them is concatenation, but a 300-dimensional learned embedding and a 2048-bit fingerprint do not contribute comparably to Euclidean distance. The fingerprint block accounts for most of it, leaving the topology learned by the encoder with little influence on the kernel.

Both failure modes arise from the feature map rather than the Gaussian process. We propose the Bounded Output Neural Dendrite (BOND), a feature map in which each output coordinate is the sum of M bounded responses weighted by range-limited coefficients. Two consequences follow. First, the range of the map is fixed by its parameterisation rather than imposed afterwards by a normalisation layer, which bounds the diameter of the feature space by a constant that does not depend on the data (Proposition 4.1) and keeps the length-scale identifiable (Theorem 4.3). Second, each output coordinate draws on both modalities through its own branches, so fusion is rebalanced at the level of individual coordinates rather than dominated by the larger input block.

The intervention is deliberately narrow. The encoder, the Gaussian process, the kernel, and the bi-level meta-learning procedure remain exactly as specified in prior work [6]. Only the map between them is changed. Our contributions are:

- We identify the geometry of the kernel’s input space as an unmanaged degree of freedom in few-shot deep-kernel learning, and demonstrate two failure modes that arise when the support set is small (Section 4).
- We propose a bounded, branch-structured feature map, prove that it bounds the diameter of the feature space by a constant independent of the data, and establish what follows for the identifiability of the kernel length-scale. The guarantee has a limit: a separation condition remains necessary, and no architectural constraint alone can provide it (Remark 4.4).
- We evaluate the method on the MoleculeNet few-shot benchmark over five random seeds on Tox21, SIDER and MUV, and at a single fixed seed on ToxCast, under both train-from-scratch and pre-trained protocols, with ablations that replace the head with bounded-but-unbranched and branched-but-unbounded alternatives.

2 Related Work

Molecular representation for few-shot learning. Two broad families of molecular representations predominate in few-shot learning. Graph-based encoders represent atoms as nodes and bonds as edges, allowing molecular topology to be learned directly [3, 13]. GIN [29], GAT, and GCN are commonly used backbones, although comparative studies suggest that the choice among them has less influence than might be expected [5, 9]. Descriptor-based encoders instead compute a fixed-length fingerprint and process it with an MLP, most commonly using ECFP [21] or RDKit descriptors [4]. Methods that combine the two representations generally concatenate them. Our starting observation is that concatenation is not a neutral operation when the resulting representation is passed to a distance-based kernel.

Meta-learning for molecular property prediction. Metric-based methods learn an embedding in which distances are meaningful, and range from Siamese networks [14] and Matching Networks [26] to ProtoNet [23] and its molecular counterpart, PAR [27]. Optimisation-based methods instead learn an initialisation that can be adapted rapidly, as exemplified by MAML [7]. ADKF-IFT [6] occupies an intermediate position. It fits a deep-kernel Gaussian process for each assay and uses the implicit function theorem to propagate gradients to the shared encoder. This framework forms the basis of our work. UniMatch [15] further introduces hierarchical attention over atoms and tasks. In contrast, we alter neither the adaptation rule nor the task construction, but instead modify the space in which the kernel operates.

Controlling feature geometry. Constraining the representation presented to a kernel or distance function is not itself new. Spectral normalisation has been used to regularise deep-kernel and distance-aware models [17, 25]. Cosine-based objectives and ℓ_2 -normalised embeddings are also standard in few-shot classification. Ober et al. [20] showed that deep kernel learning can overfit by deforming the feature space as the marginal likelihood is maximised. To our knowledge, however, prior work has not examined this deformation when the support set contains only twenty points and the kernel hyper-parameters are refitted for every assay. This is precisely the regime encountered in few-shot molecular property

prediction. We address this gap, and our ablations therefore include ℓ_2 normalisation and a bounded single-layer map as substantive controls rather than strawmen.

3 Method

Figure 3 summarises the overall pipeline.

Problem setting. Let $\mathcal{T}$ denote a distribution over binary assays. A task $T \sim \mathcal{T}$ provides a support set $S_T = \{(x_i, y_i)\}_{i=1}^K$ and a query set $Q_T = \{(x_j, y_j)\}_{j=1}^Q$, where x denotes a molecule and $y \in \{0, 1\}$ is its label. We adopt the standard 2-way 10-shot protocol, for which $|S_T| = 20$. The objective is to learn a predictor that, conditioned on S_T , performs accurately on Q_T .

Encoder (inherited). Molecular topology is encoded by a five-layer GIN [29] with mean pooling, giving a 300-dimensional graph embedding h_G ; in parallel each molecule is hashed to a 2048-bit ECFP4 fingerprint $f(x)$. The two are concatenated into a 2348-dimensional vector. The architecture and the optional pre-trained initialisation follow prior work [6, 12], and Appendix H gives the configuration in full.

Bounded branch head. The concatenated vector is mapped to $z \in \mathbb{R}^d$, $d = 2048$, according to

$$z_j = \sum_{m=1}^M \underbrace{\tanh(w_{jm})}_{\text{branch weight}} \underbrace{\varsigma(a_{jm}^\top x + b_{jm})}_{\text{bounded response}}, \quad j = 1, \dots, d, \quad (1)$$

where ς is the logistic function and $M = 30$ is the number of branches. The form is loosely motivated by dendritic integration, in which a neuron aggregates locally saturating inputs along separate branches [8]; the aggregation in (1) is additive, unlike the multiplicative aggregation of the biological model. The right panel of Figure 3 illustrates the module.

The architecture imposes two structural properties rather than learning them. First, each response lies in $(0, 1)$ and each branch weight in $(-1, 1)$, so z_j stays within $(-M, M)$ whatever the encoder does; Proposition 4.1 turns this coordinate-wise constraint into a bound on the diameter of the whole feature space. Second, the map from the 2348-dimensional input to the M responses of coordinate j is learned jointly over both modalities rather than applied to each separately. We are careful about the scope of that second property: Section 5.6 measures each block’s contribution to the output distance and finds the fingerprint still dominant after the head, so the joint parameterisation is the mechanism by which both views reach the kernel at all, not a guarantee that their scales are equalised.

Gaussian process and meta-learning (inherited). Following [6], z is fed to a Gaussian process with zero mean and the Matérn-5/2 kernel of Eq. (3), whose log marginal likelihood is given in Eq. (13). Training is bi-level: the encoder and head are shared across assays, while the three kernel hyper-parameters $\psi = (\log \ell, \log s, \log \sigma_n^2)$ are refitted for every assay by L-BFGS on its support set, starting from the median heuristic — a choice Corollary 4.5 justifies for this kernel. The outer loop evaluates the predictive loss on the query set and, rather than unrolling the inner

optimisation, obtains the hypergradient from the implicit function theorem:

$$\nabla_{\phi} \mathcal{L}_{\text{val}}^t = -\nabla_{\phi} \nabla_{\psi} J_t [\nabla_{\psi}^2 J_t]^{-1} \nabla_{\psi} \mathcal{L}_{\text{val}}^t. \quad (2)$$

Equation (2) is what makes the geometry of z an operational concern rather than an aesthetic one: it inverts $\nabla_{\psi}^2 J_t$, and Theorem 4.3 is a statement about when the $(\log \ell, \log \ell)$ entry of that matrix stays away from zero.

4 Why a Bounded Feature Map Helps

We formalise the argument of Section 1. Throughout, we consider a support set of n molecules mapped to features $z_i \in \mathbb{R}^d$ and evaluated using a Matérn-5/2 kernel with a single shared length-scale:

$$K_{ij} = s k(r_{ij}), \quad r_{ij} = \frac{\|z_i - z_j\|_2}{\ell}, \quad (3)$$

where $k(r) = (1 + \sqrt{5}r + \frac{5}{3}r^2)e^{-\sqrt{5}r}$, $s > 0$ is the output scale, and σ_n^2 is the Gaussian noise variance. The inner problem estimates $\psi = (\log \ell, \log s, \log \sigma_n^2)$ by maximising the log marginal likelihood $J(\psi)$. The outer problem differentiates through this solution using the implicit function theorem, which requires $\nabla_{\psi}^2 J$ to be invertible.

The head imposes a data-independent range. Coordinate-wise, the proposed head is $h_j(x) = \sum_{m=1}^M A_{jm} \zeta(a_{jm}^T x + b_{jm})$, where ζ is the logistic function and $A_{jm} = \tanh(w_{jm})$.

Proposition 4.1 (Structural range bound). *For every pair of inputs x, x' and every coordinate j , $|h_j(x) - h_j(x')| \leq \|A_{j\cdot}\|_1$, and consequently*

$$\text{diam } h(X) \leq R_{\star} := \left(\sum_{j=1}^d \|A_{j\cdot}\|_1^2 \right)^{1/2} \leq M\sqrt{d}. \quad (4)$$

$R_{\star}$ depends only on the gate weights, not on the data, the input distribution, or the preceding encoder. A linear or ReLU head admits no such constant: its diameter is $\|V\|_{\text{op}} \text{diam}(X)$, which the optimiser can increase without constraint during meta-training.

Length-scale information vanishes at both extremes. Differentiating (3) yields $\partial_{\log \ell} K_{ij} = s g(r_{ij})$, where

$$g(r) := -r k'(r) = \frac{5}{3} r^2 (1 + \sqrt{5}r) e^{-\sqrt{5}r} \geq 0. \quad (5)$$

Proposition 4.2 (Information profile). *g is unimodal on $(0, \infty)$ with a unique maximiser $r^{\star} = (1 + \sqrt{3})/\sqrt{5} \approx 1.2218$ and $g(r^{\star}) \approx 0.6044$. Moreover, $g(r) = \Theta(r^2)$ as $r \rightarrow 0$ and $g(r) = \Theta(r^3 e^{-\sqrt{5}r})$ as $r \rightarrow \infty$.*

The length-scale therefore becomes unidentifiable in two limiting regimes: as the features collapse, $r \rightarrow 0$ and $K \rightarrow s\mathbf{1}\mathbf{1}^T$; as they disperse, $r \rightarrow \infty$ and $K \rightarrow sI$. In either case $\partial_{\log \ell} K \rightarrow 0$ and the curvature of J along $\log \ell$ vanishes.

Theorem 4.3 (Non-degeneracy of the inner problem). *Suppose the support features are q -separated, $\min_{i \neq j} \|z_i - z_j\| \geq q > 0$. Then the Fisher information of J in the $\log \ell$ direction obeys*

$$\mathcal{I}_{\ell\ell} \geq \frac{n(n-1)s^2 g^2}{2(ns + \sigma_n^2)^2}, \quad \underline{g} = \min\{g(q/\ell), g(R_{\star}/\ell)\} > 0, \quad (6)$$

where $R_{\star}$ is the constant defined in Proposition 4.1.

The theorem bounds one diagonal entry of the expected Hessian away from zero, which is necessary for the inner problem to be non-degenerate along $\log \ell$. Invertibility of the realised 3×3 Hessian used in (2) does not follow from it and is not claimed here. The bound needs two ingredients: an upper bound on the diameter, which the architecture supplies, and a lower bound on the separation, which it does not.

Remark 4.4 (What boundedness does *not* buy). Proposition 4.1 alone is insufficient, and the gap is not an artefact of the proof. The constant map $h \equiv z_0$ satisfies (4) but has $q = 0$, so $r_{ij} \equiv 0$, $g \equiv 0$ and $\mathcal{I}_{\ell\ell} = 0$: the kernel matrix is rank one and the length-scale is completely unidentifiable. Boundedness prevents dispersion but not collapse. We therefore treat separation as a property to be verified rather than assumed, and measure it in Appendix G.

Corollary 4.5 (The median heuristic is nearly information-optimal). *Initialising $\ell_0 = (\frac{1}{2} \text{median}_{i < j} \|z_i - z_j\|)^{1/2}$ places the median pair at $r = \sqrt{2}$. For the Matérn-5/2 kernel, $g(\sqrt{2})/g(r^{\star}) = 0.9718$, so the typical support pair sits at 97.2% of the peak of the information profile g , and hence at $0.9718^2 = 94.4\%$ of the maximum available Fisher information, which is proportional to g^2 .*

Corollary 4.5 gives a theoretical basis for a heuristic the deep-kernel literature has adopted without formal justification, and Theorem 4.3 states when that justification survives a feature map that is itself learned. Proofs are in Appendix F.

5 Experiments

5.1 Setup

We evaluate on MoleculeNet [28] using the few-shot split introduced by PAR [27] and adopted by ADKF-IFT [6], so that our numbers are directly comparable with published ones. Each dataset is a panel of binary assays split into disjoint meta-training and meta-testing sets; Table 4 gives the sizes. We use the standard 2-way 10-shot protocol, with a support set of 20 molecules and a query set of 16, and report AUROC averaged over the meta-test assays. Appendix H gives the full configuration, compute and ablation arms.

Two caveats on how a number should be read. First, following [6, 27] the reported figure is the *highest* mean meta-test AUROC across a run's evaluation rounds. Because a maximum is optimistically biased, Appendix M reports the final-round value for every cell; the two conventions should be compared separately rather than mixed. Second, MUV saturates near 100 within the first few evaluation rounds for every feature map we tried, including controls the theory predicts should degenerate (Figure 4c). A saturated metric cannot separate methods, so we draw no conclusion from the MUV column and read the ablations off Tox21 and SIDER. A control in Appendix L traces the saturation to meta-test query labels reaching the per-task adaptation step: permuting them drops MUV from 99.97 to 79.79, and removing the adaptation entirely drops it to 72.13. This is a property of the benchmark protocol we adopted rather than of any method, and we leave the evaluation path unmodified so that our numbers stay comparable with published ones.

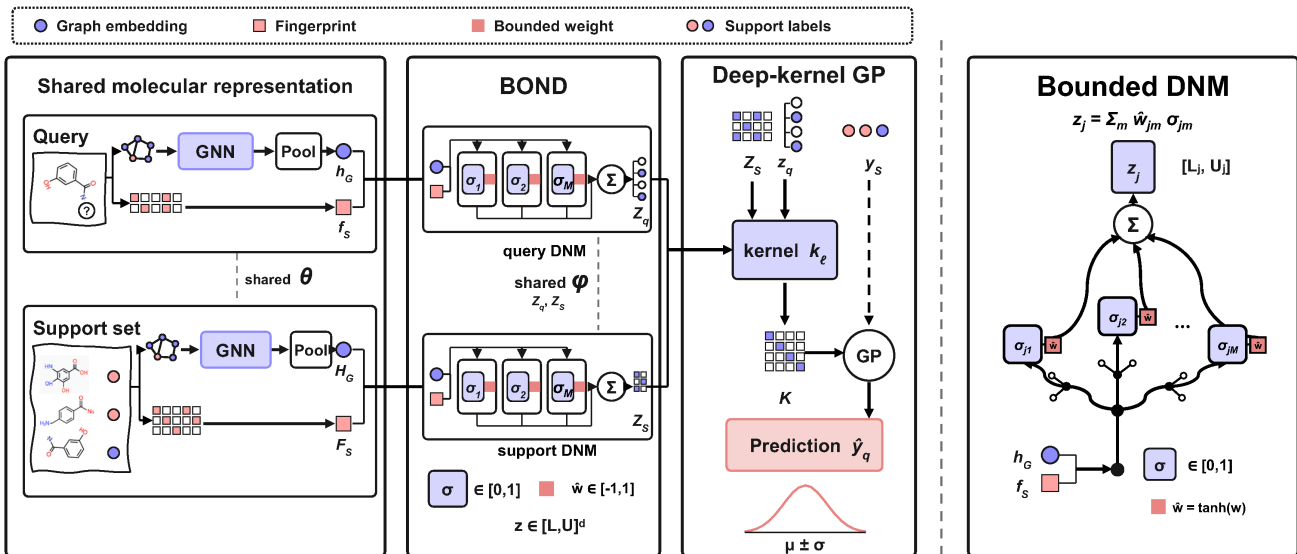


Figure 3: Overview of BOND; the head is labelled DNM (dendritic neuron module) inside the panels. Left. A shared encoder maps every molecule to a graph embedding and an ECFP4 fingerprint; the head turns that pair into z , and the deep-kernel Gaussian process conditions on the support set to predict the query. The encoder and head parameters θ, ϕ are shared across assays, while the kernel hyper-parameters are refitted for each one. Right. The head itself: each output coordinate sums M bounded responses through range-limited weights, which confines z to a box the data cannot enlarge.

Repetitions and baselines. On Tox21, SIDER and MUV every BOND cell is run with five seeds, 0 through 4, and we report the mean and population standard deviation over them. The seed jointly controls assay split ordering, episode sampling and initialisation. ToxCast is instead repeated at a single fixed seed, so its interval measures run-to-run non-determinism rather than seed variation and is not comparable with the others. Baseline numbers are quoted from the original publications as collected in [6] and were not re-run, so their error bars follow each source paper’s own repetition scheme. We compare against metric-based learners (Siamese [14], ProtoNet [23], TPN [18], EGNN [13], PAR [27]), optimisation-based learners (MAML [7], IterRefLSTM [3], ADKF-IFT [6], UniMatch [15]), and graph or hybrid models (MixHop [1], CHEF [2], MHNfs [22]), under both protocols.

Ablations. The ablations change the head and nothing else, so any difference is attributable to the feature map. We compare the proposed head with a branched but unbounded head, a bounded but unbranched head $\tanh(Wx)$, an ℓ_2 -normalised linear map, and a two-layer MLP. The middle two test whether normalisation alone accounts for any benefit. The MLP is matched on *parameter count* rather than width, because the proposed head holds 98.7% of the model’s parameters and a width-matched control would confound architecture with capacity. We also sweep the number of branches M .

5.2 Main results

Table 1 compares BOND with the published baselines; Figure 8 in the appendix plots the same comparison as a gain over the strongest

baseline on each panel. From scratch, BOND leads on Tox21 (89.78 against 82.62) and SIDER (89.56 against 74.68) by 7.16 and 14.88 AUROC; pre-trained, by 2.66 and 8.43 (89.01 against 86.35, 88.77 against 80.34). ToxCast is the one panel where it does not lead (76.80 against 77.74; 78.51 against 81.63), and MUV is not interpreted, for the reason given in Section 5.1. The Avg. margins of +10.33 and +8.24 are inflated by the saturated MUV entry and are not a headline result; Section 6.1 returns to the ToxCast interval.

5.3 Ablation of the feature map

Table 2 compares the proposed head with alternative feature maps. Three findings emerge, presented in increasing order of tension with our initial hypothesis.

First, the differences among heads are modest. The proposed head performs best on Tox21 and SIDER, exceeding the branched unbounded map by 0.66 and 0.48 AUROC, respectively. These margins should be considered alongside seed-to-seed standard deviations of 0.37 and 0.26. Every configuration that remains trainable falls within 2.7 AUROC points of the others. The evidence therefore supports robustness to the choice of feature map more strongly than it supports a substantial improvement attributable to the proposed head.

Second, two variants terminate before reaching the epoch budget because the implicit-function-theorem solve becomes singular. Their outcomes are nevertheless qualitatively different. The parameter-matched MLP terminates between epochs 70–340, having failed to train beyond scores of 63.0 on Tox21 and 56.9 on SIDER. The single-branch $\tanh$ head terminates much later, between epochs 1420–1920 of 2000, after converging to scores of 87.1 and 88.0. The

Table 1: Few-shot performance on MoleculeNet (2-way 10-shot, meta-test AUROC %). Baselines are quoted from their original publications. Avg. is over the three panels every method reports; best per column in red, second in blue.

Method	Tox21(8,014/12)↑	SIDER(1,427/27)↑	MUV(93,127/17)↑	ToxCast(8,615/617)↑	Avg.
Siamese [14]	80.40 ± 0.35	71.10 ± 4.32	59.59 ± 5.13	–	70.36
ProtoNet [23]	74.98 ± 0.32	64.54 ± 0.89	65.88 ± 4.11	63.70 ± 1.26	68.47
MAML [7]	80.21 ± 0.24	70.43 ± 0.76	63.90 ± 2.28	66.79 ± 0.85	71.51
IterReflSTM [3]	81.10 ± 0.17	69.63 ± 0.31	45.56 ± 5.12	–	65.43
TPN [18]	76.05 ± 0.24	67.84 ± 0.95	65.22 ± 5.82	62.74 ± 1.45	69.70
EGNN [13]	81.21 ± 0.16	72.87 ± 0.73	65.20 ± 2.08	63.65 ± 1.57	73.09
MixHop [1]	78.14 ± 0.33	72.01 ± 0.87	78.04 ± 3.01	77.19 ± 0.93	76.06
CHEF [2]	61.97 ± 0.65	57.34 ± 0.82	53.17 ± 4.21	56.52 ± 1.24	57.49
PAR [27]	82.06 ± 0.12	74.68 ± 0.31	66.48 ± 2.12	69.72 ± 1.63	74.41
ADKF-IFT [6]	82.43 ± 0.60	67.72 ± 1.21	98.18 ± 3.05	72.07 ± 0.81	82.78
MHNFs [22]	80.23 ± 0.84	65.89 ± 1.17	73.81 ± 2.53	74.91 ± 0.73	73.31
UniMatch [15]	82.62 ± 0.43	68.13 ± 1.54	79.40 ± 3.14	77.74 ± 0.75	76.72
BOND (Ours)	89.78 ± 0.37	89.56 ± 0.26	99.99 ± 0.00	76.80 ± 0.12	93.11
Pre-GNN [12]	82.14 ± 0.08	73.96 ± 0.08	67.14 ± 1.58	73.68 ± 0.74	74.41
GNN-MAML [10]	82.97 ± 0.10	75.43 ± 0.21	68.99 ± 1.84	–	75.80
Pre-PAR [27]	84.93 ± 0.11	78.08 ± 0.16	69.96 ± 1.37	75.12 ± 0.84	77.66
Pre-ADKF-IFT [6]	86.06 ± 0.35	70.95 ± 0.60	95.74 ± 0.37	76.22 ± 0.13	84.25
MolFeSCue [30]	85.93 ± 0.10	79.08 ± 0.14	72.96 ± 1.18	76.39 ± 1.52	79.32
Pre-UniMatch [15]	86.35 ± 0.13	80.34 ± 0.45	86.35 ± 0.76	81.63 ± 0.73	84.35
Pre-BOND (Ours)	89.01 ± 0.23	88.77 ± 0.30	99.98 ± 0.00	78.51 ± 0.01	92.59

pattern across variants aligns more closely with branch structure than with boundedness. The bounded tanh head terminates before completion, whereas the branched unbounded head completes all 9/9 runs. Across these ablations, successful completion is more closely associated with the branched component of the architecture than with squashing alone.

Third, the final row provides the strongest challenge to our initial framing. This variant removes the fingerprint block rather than changing the head, reducing AUROC by 16.3 on Tox21, 30.7 on SIDER, and 12.7 on MUV. These reductions are an order of magnitude larger than those caused by any head substitution. The fingerprint dominance quantified in Section 5.6 is therefore not a pathology corrected by the head. Instead, the fingerprint appears to contain a substantial fraction of the predictive signal. The second premise of Section 1 should accordingly be interpreted as motivation for managing the geometry of the fused space, rather than as a defect that the proposed architecture eliminates.

5.4 Sensitivity to the number of branches

The head is Linear(2348 → 2048M), so M determines most of the model capacity and its parameter count grows linearly. Performance is not monotonic in M on either dataset; Appendix I gives the sweep, together with the capacity cost and a note on its provenance.

5.5 Convergence

Figure 4 provides a view that aggregate means cannot give. It shows the meta-training trajectories of the three feature maps that complete training, reporting the median across seeds with an interquartile band. The ℓ_2 -normalised head improves markedly faster over

Table 2: Head ablation, trained from scratch (best AUROC %), with everything outside the feature map held fixed. † marks arms whose runs aborted before the epoch budget; the value is the score reached before the abort (see text).

Feature map	Tox21	SIDER	MUV
BOND (bounded, branched)	89.78 ± 0.37	89.56 ± 0.26	99.99 ± 0.00
branched, unbounded	89.12 ± 0.29	89.08 ± 0.19	99.99 ± 0.00
ℓ_2 -norm, 1 branch	88.32 ± 0.31	88.72 ± 0.45	99.99 ± 0.00
tanh, 1 branch	87.05 ± 0.14 [†]	87.99 ± 0.15 [†]	99.98 ± 0.00 [†]
MLP, matched	63.00 ± 1.30 [†]	56.86 ± 0.36 [†]	99.99 ± 0.00
BOND , no fingerprint	73.46 ± 0.69	58.84 ± 0.78	87.29 ± 0.52

the first few hundred epochs on both Tox21 and SIDER, by up to fifteen AUROC points, and is overtaken at roughly 900 epochs on Tox21 and 1400 on SIDER; on MUV the trajectories are close throughout.

The panel averages also conceal substantial per-assay heterogeneity. Performance varies by more than ten AUROC points across the six SIDER assays and by nearly fifty across the 156 ToxCast assays, so a panel mean does not characterise the behaviour on any individual assay. Figure 10 in the appendix gives the full per-assay breakdown.

5.6 What happens to the feature geometry

Section 4 names two modes in which the kernel’s input space can degenerate. Both are measurable, so we evaluate them on saved checkpoints rather than inferring them from AUROC.

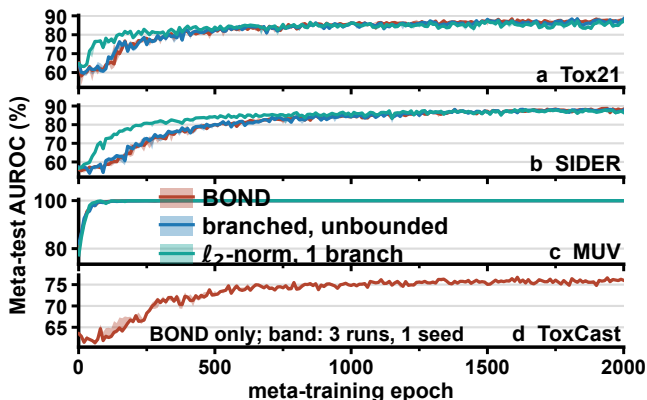


Figure 4: Meta-training trajectories for each dataset. Curves show the median across runs, and shaded bands denote the interquartile range.

Figure 5a reports the diameter of the mapped support set before and after meta-training. The unbounded head lets it grow by a factor of 95 on SIDER and 23 on Tox21, the $r \rightarrow \infty$ regime in which the marginal likelihood can rescale the geometry instead of improving it. The ℓ_2 -normalised head fails in the opposite direction, contracting to 0.09 on SIDER and 0.02 on MUV, which approaches the $r \rightarrow 0$ regime in which every pair sits at the same distance and the length-scale stops being identifiable. These are the two limits of Proposition 4.2, observed rather than assumed.

Two qualifications follow. First, every head whose degeneration we can watch is one that trains to completion; we logged neither $\mathcal{I}_{\ell\ell}$ nor the null direction of the inner Hessian for the arms that abort, so the theory is consistent with those failures rather than shown to cause them. Second, the proposed head drifts too, by a factor of 24 on SIDER, so R_* is far too loose to be the stabilising mechanism on its own; on MUV it instead contracts to a diameter of 1.1×10^{-4} , further into the $r \rightarrow 0$ regime than the ℓ_2 -normalised head it is meant to improve on, while still scoring 99.99 – one more reason to read nothing from that panel. Boundedness changes the rate at which the geometry deforms; it does not hold it fixed.

Figure 5b evaluates the second premise. Splitting the squared pairwise distance at the head’s input into its graph and fingerprint blocks gives the fingerprint 83% on Tox21, 85% on ToxCast, 89% on SIDER and 92% on MUV, so concatenation hands the metric to the fingerprint on every panel. Measuring each block’s influence *after* the head, by zeroing one and recomputing, leaves the output distance correlating 0.96 with the fingerprint alone and 0.10 with the graph alone on SIDER. The head therefore does not equalise the two views and we do not claim it does; the graph reaches the kernel further on the larger panels, at 0.36 on Tox21 and 0.28 on ToxCast, but nowhere does it rival the fingerprint.

Figure 6 answers a question the diameter alone cannot: whether the geometry that moves is geometry the kernel can see. Panel (a) makes the change of scale visible, with scale bars four orders of magnitude apart. Panel (b) is the control. Because the length-scale is initialised from the median pairwise distance, a uniform rescaling is divided out exactly and the kernel matrix is unchanged; normalising

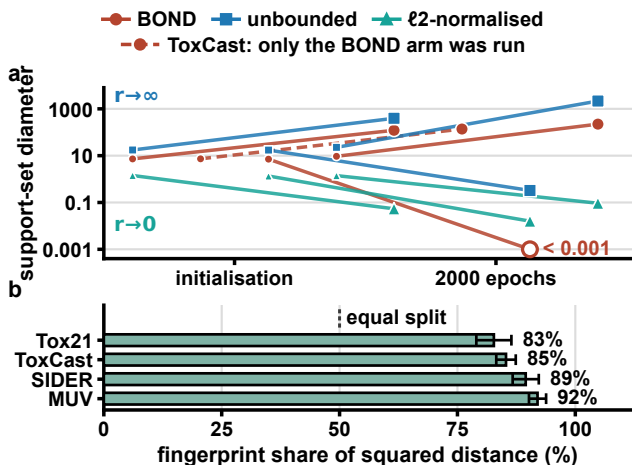


Figure 5: Measured feature geometry. (a) Diameter of the mapped support set before and after meta-training, shown on a logarithmic axis. (b) Proportion of the squared pairwise distance at the input to the head contributed by the fingerprint block.

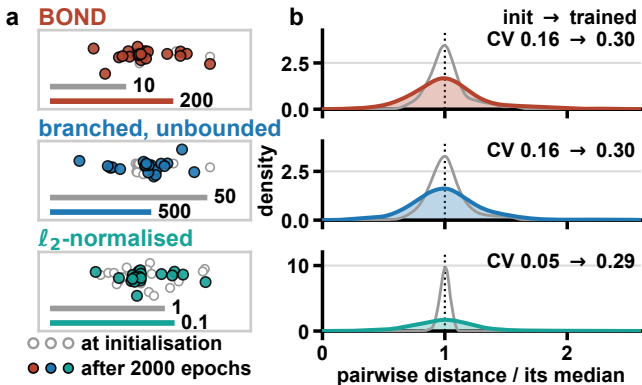


Figure 6: Deformation of the feature geometry. (a) Support molecules projected onto their first two principal components before (open) and after (filled) meta-training. (b) The same pairwise distances after division by their respective medians.

each distance by its own median therefore leaves only genuine reshaping. Doing so, the three heads converge: their coefficients of variation begin at 0.16, 0.16 and 0.05 and end at 0.30, 0.30 and 0.29. The four orders of magnitude separating the raw diameters are almost entirely absorbed, and what reaches the kernel is close to identical.

We take two things from this. The median heuristic is scale-equivariant, so it divides out exactly the quantity Proposition 4.1 bounds, and it is the most plausible explanation for three heads with very different raw geometries landing within one AUROC point of each other. But it fixes the scale of the distance distribution, not its shape, and cannot recover a length-scale gradient once the distances carry no information about it. We stop short of crediting

the bound with preventing the observed failures, because the failure pattern does not support it: the bounded single-branch head aborts and the unbounded branched head does not.

5.7 What the feature space looks like

Because our central claim concerns the geometry presented to the kernel, we also inspect that geometry directly, projecting the mapped molecules of a Tox21 meta-test assay and reading off the atom-level weights the head assigns to a single molecule. Under BOND the two classes begin to occupy distinguishable regions, whereas under the baseline head they remain more intermingled. The attribution is more equivocal: BOND with $M = 30$ recovers one of the two substructures we regard as informative for this endpoint and the baseline recovers neither, but all three models place most of their weight on backbone carbons. Both views are qualitative — a two-dimensional projection of a 2048-dimensional space is suggestive rather than conclusive, and the attribution panel is one molecule from one assay — so we give them in Appendix O, beside the larger populations they were drawn from.

6 Conclusion

Few-shot deep-kernel methods assign much of the burden of generalisation to the kernel, whose effectiveness depends critically on the geometry of the space in which distances are measured. We have argued that this geometry has remained largely unmanaged and exhibits two distinct failure modes in the small-support regime: an unbounded map permits rescaling that the marginal likelihood cannot penalise, while direct concatenation allows the fingerprint to dominate the graph representation. The proposed bounded branch head imposes a data-independent upper bound on the diameter of the feature space, ruling out the dispersion regime in which the length-scale becomes unidentifiable. On Tox21 and SIDER, the proposed method outperforms the strongest published baselines under both training protocols.

Our empirical analysis also places three important qualifications on this account. First, the median-heuristic initialisation is exactly scale-equivariant and therefore removes the absolute scale constrained by the bound. After normalisation by the median distance, heads whose raw diameters differ by four orders of magnitude present nearly identical relative geometries to the kernel, and their AUROC values differ by less than one point. Second, the proposed head does not equalise the two modalities. The fingerprint remains the principal contributor to the output distance, and removing it incurs an order of magnitude greater loss in accuracy than any change to the head. Third, the variants that terminate prematurely are not those for which we directly observe geometric degeneration. The theoretical mechanism is therefore consistent with these failures but is not established as their cause by the present experiments.

6.1 Limitations and Ethical Considerations

Five considerations delimit the conclusions that can be drawn from these results. Appendix N provides the supporting measurements and identifies the experiment needed to resolve most of these. First, the MUV column is uninformative because the metric saturates for every variant. A diagnostic control attributes this saturation to meta-test query labels entering the per-task adaptation step

(Appendix L). Second, the ToxCast experiments are complete but use a single fixed seed. Their intervals therefore capture run-to-run non-determinism rather than variation across seeds, and ToxCast is the only dataset on which the proposed head does not lead. Third, the diameter bound is expressed in terms of R_* , which lies two orders of magnitude above the realised geometry. Consequently, the Fisher-information guarantee is numerically negligible over the range in which it is evaluated. Fourth, we ablate the fingerprint but not the encoder, leaving a fingerprint-only model as an important missing control. Finally, our analysis assumes an isotropic stationary kernel with a single length-scale. An anisotropic or deep-kernel variant would make a different geometric quantity relevant.

On ethics, this work uses only public molecular property benchmarks, involves no human or animal subjects and generates no personal data, and it satisfies the research ethics requirements of the University of Toyama, Ritsumeikan University, Sichuan University and Tongji University. One dual-use consideration is common to the area: a toxicity model can in principle be inverted to search for harmful compounds. The accuracy reported here is far below what that would require and the benchmarks are long-standing public resources, so we judge the marginal risk negligible.

7 Generative AI Usage

Generative AI tools were used solely to correct English grammar in prose the authors had already written. They were not used to generate research ideas, to design or run experiments, or to produce or alter any datum, figure or table. Every number and claim in this paper comes from the experiments described here, and the authors take full responsibility for all of them.

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References

- [1] Sami Abu-El-Hajja, Bryan Perozzi, Amol Kapoor, Nazanin Alipourfard, Kristina Lerman, Hrayr Harutyunyan, Greg Ver Steeg, and Aram Galstyan. 2019. MixHop: Higher-order graph convolutional architectures via sparsified neighborhood mixing. In *International Conference on Machine Learning*. 21–29.
- [2] Thomas Adler, Johannes Brandstetter, Michael Widrich, Andreas Mayr, David Kreil, Michael Kopp, Günter Klambauer, and Sepp Hochreiter. 2020. Cross-domain few-shot learning by representation fusion. *arXiv preprint arXiv:2010.06498* (2020).
- [3] Han Altae-Tran, Bharath Ramsundar, Aneesh S Pappu, and Vijay Pande. 2017. Low data drug discovery with one-shot learning. *ACS Central Science* 3, 4 (2017), 283–293.
- [4] A Patricia Bento, Anne Hersey, Eloy Félix, Greg Landrum, Anna Gaulton, Francis Atkinson, Louisa J Bellis, Marleen De Veij, and Andrew R Leach. 2020. An open source chemical structure curation pipeline using RDKit. *Journal of Cheminformatics* 12, 1 (2020), 51.
- [5] Deli Chen, Yankai Lin, Wei Li, Peng Li, Jie Zhou, and Xu Sun. 2020. Measuring and relieving the over-smoothing problem for graph neural networks from the topological view. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34. 3438–3445.
- [6] Wenlin Chen, Austin Tripp, and José Miguel Hernández-Lobato. 2023. Meta-learning adaptive deep kernel Gaussian processes for molecular property prediction. In *International Conference on Learning Representations*.

- [7] Chelsea Finn, Pieter Abbeel, and Sergey Levine. 2017. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning*. 1126–1135.
- [8] Shangce Gao, Mengchu Zhou, Yirui Wang, Jiuju Cheng, Hanaki Yachi, and Jiahai Wang. 2019. Dendritic neuron model with effective learning algorithms for classification, approximation, and prediction. *IEEE Transactions on Neural Networks and Learning Systems* 30, 2 (2019), 601–614.
- [9] Justin Gilmer, Samuel S Schoenholz, Patrick F Riley, Oriol Vinyals, and George E Dahl. 2017. Neural message passing for quantum chemistry. In *International Conference on Machine Learning*. 1263–1272.
- [10] Zhichun Guo, Chuxu Zhang, Wenhao Yu, John Herr, Olaf Wiest, Meng Jiang, and Nitesh V. Chawla. 2021. Few-shot graph learning for molecular property prediction. In *Proceedings of the Web Conference*. 2559–2567.
- [11] Rohan Gupta, Devesh Srivastava, Mehar Sahu, Swati Tiwari, Rashmi K Ambasta, and Pravir Kumar. 2021. Artificial intelligence to deep learning: machine intelligence approach for drug discovery. *Molecular Diversity* 25, 3 (2021), 1315–1360.
- [12] Weihua Hu, Bowen Liu, Joseph Gomes, Marinka Zitnik, Percy Liang, Vijay Pande, and Jure Leskovec. 2020. Strategies for pre-training graph neural networks. In *International Conference on Learning Representations*.
- [13] Jongmin Kim, Taesup Kim, Sungwoong Kim, and Chang D Yoo. 2019. Edge-labeling graph neural network for few-shot learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11–20.
- [14] Gregory Koch, Richard Zemel, and Ruslan Salakhutdinov. 2015. Siamese neural networks for one-shot image recognition. In *ICML Deep Learning Workshop*, Vol. 2.
- [15] Ruifeng Li, Mingqian Li, Wei Liu, Yuhua Zhou, Xiangxin Zhou, Yuan Yao, Qiang Zhang, and Hongyang Chen. 2025. UniMatch: Universal matching from atom to task for few-shot drug discovery. In *International Conference on Learning Representations*.
- [16] Xiaoxu Li, Xiaochen Yang, Zhanyu Ma, and Jing-Hao Xue. 2023. Deep metric learning for few-shot image classification: A review of recent developments. *Pattern Recognition* 138 (2023), 109381.
- [17] Jeremiah Zhe Liu, Zi Lin, Shreyas Padhy, Dustin Tran, Tania Bedrax-Weiss, and Balaji Lakshminarayanan. 2020. Simple and principled uncertainty estimation with deterministic deep learning via distance awareness. In *Advances in Neural Information Processing Systems*, Vol. 33. 7498–7512.
- [18] Yanbin Liu, Juho Lee, Minseop Park, Saehoon Kim, Eunho Yang, Sung Ju Hwang, and Yi Yang. 2019. Learning to propagate labels: Transductive propagation network for few-shot learning. In *International Conference on Learning Representations*.
- [19] Jianzhu Ma, Samson H Fong, Yunan Luo, Christopher J Bakkenist, John Paul Shen, Soufiane Mourragui, Lodewyk F A Wessels, Marc Hafner, Roded Sharan, Jian Peng, and Trey Ideker. 2021. Few-shot learning creates predictive models of drug response that translate from high-throughput screens to individual patients. *Nature Cancer* 2, 2 (2021), 233–244.
- [20] Sebastian W Ober, Carl E Rasmussen, and Mark van der Wilk. 2021. The promises and pitfalls of deep kernel learning. In *Uncertainty in Artificial Intelligence*, Vol. 161. 1206–1216.
- [21] David Rogers and Mathew Hahn. 2010. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling* 50, 5 (2010), 742–754.
- [22] Johannes Schimunek, Philipp Seidl, Lukas Friedrich, Daniel Kuhn, Friedrich Rippmann, Sepp Hochreiter, and Günter Klambauer. 2023. Context-enriched molecule representations improve few-shot drug discovery. In *International Conference on Learning Representations*.
- [23] Jake Snell, Kevin Swersky, and Richard Zemel. 2017. Prototypical networks for few-shot learning. In *Advances in Neural Information Processing Systems*, Vol. 30.
- [24] Alexander Tropsha, Olexandr Isayev, Alexandre Varnek, Gisbert Schneider, and Artem Cherkasov. 2024. Integrating QSAR modelling and deep learning in drug discovery: the emergence of deep QSAR. *Nature Reviews Drug Discovery* 23, 2 (2024), 141–155.
- [25] Joost van Amersfoort, Lewis Smith, Andrew Jesson, Oscar Key, and Yarin Gal. 2021. On feature collapse and deep kernel learning for single forward pass uncertainty. *arXiv preprint arXiv:2102.11409* (2021).
- [26] Oriol Vinyals, Charles Blundell, Timothy Lillicrap, Koray Kavukcuoglu, and Daan Wierstra. 2016. Matching networks for one shot learning. In *Advances in Neural Information Processing Systems*, Vol. 29. 3630–3638.
- [27] Yaqing Wang, Abulikemu Abuduweili, Quanming Yao, and Dejing Dou. 2021. Property-aware relation networks for few-shot molecular property prediction. In *Advances in Neural Information Processing Systems*, Vol. 34. 17441–17454.
- [28] Zhenqin Wu, Bharath Ramsundar, Evan N Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S Pappu, Karl Leswing, and Vijay Pande. 2018. MoleculeNet: a benchmark for molecular machine learning. *Chemical Science* 9, 2 (2018), 513–530.
- [29] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. 2019. How powerful are graph neural networks?. In *International Conference on Learning Representations*.
- [30] Ruochi Zhang, Chao Wu, Qian Yang, Chang Liu, Yan Wang, Kewei Li, Lan Huang, and Fengfeng Zhou. 2024. MolFeSCue: enhancing molecular property prediction in data-limited and imbalanced contexts using few-shot and contrastive learning. *Bioinformatics* 40, 4 (2024), btac118.
- [31] Haiteng Zhao, Shengchao Liu, Chang Ma, Hannan Xu, Jie Fu, Zhi-Hong Deng, Lingpeng Kong, and Qi Liu. 2023. GIMLET: A unified graph-text model for instruction-based molecule zero-shot learning. In *Advances in Neural Information Processing Systems*, Vol. 36. 5850–5887.
- [32] Yangqing Zhong, Yuling Su, and Hong Zhao. 2023. Self-similarity feature based few-shot learning via hierarchical relation network. *International Journal of Machine Learning and Cybernetics* 14, 12 (2023), 4237–4249.

Supplementary Material for “What the Kernel Measures: Bounding Feature Geometry for Few-Shot Molecular Property Prediction”

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Table 3: Configuration. These are the shipped defaults; only the seed and the number of epochs were varied across the runs reported here.

Setting	Value
Shots per class (train / test)	10 / 10
Query size	16
Meta-batch (assays per step)	9
Inner / outer steps per episode	1 / 1
Meta-training epochs	2000
Evaluation interval	every 10 epochs
Seeds	0, 1, 2, 3, 4
Encoder	GIN, 5 layers
Embedding dimension	300
Readout / pooling	last layer / mean
Dropout	0.5
Fingerprint	ECFP4, 2048 bits
Branches M	30
Head output dimension d	2048
Base kernel	Matérn-5/2, isotropic
Length-scale initialisation	median heuristic
Noise prior	log-normal, mode 0.1
Inner solver	L-BFGS on the log marginal likelihood
Outer optimiser	AdamW, lr 10^{-3} , wd 5×10^{-5}
Gradient clipping	1.0

A The Meta-Training Procedure

Algorithm 1 is the loop that produced every BOND number in this paper. Two notational points. We write ϕ for the parameters shared across assays — the GIN encoder together with the head weights $\{a_{jm}, b_{jm}, w_{jm}\}$ — to match Eq. (2), and $u(x)$ for the concatenated encoding, which Section F.1 calls $\phi(x)$; the two are kept apart here because both appear in the same lines. And $\psi = (\log \ell, \log s, \log \sigma_n^2)$ is the entire set of variables the inner loop touches: three scalars, refitted from scratch for every task, discarded afterwards. No state carries from one task to the next except through ϕ .

The bi-level structure, the episode construction and the hyper-gradient are inherited unchanged from ADKF-IFT [6], which in turn follows the task construction of PAR [27]; line 7 is the only line this paper changes. The asymmetry between the loops is the point of the design. The inner problem is small enough to solve to optimality — a three-parameter concave-in-practice fit on twenty points — so the outer loop can differentiate through its solution rather than through an unrolled optimiser trajectory, which is what makes line 11 cheap. It also makes line 11 fragile: it requires $\nabla_{\psi}^2 J_t$ to be invertible, and Appendix B shows exactly where that requirement enters. Theorem 4.3 is a statement about when the $(\log \ell, \log \ell)$

Algorithm 1 Meta-training BOND

Input: meta-training assays $\mathcal{T}$; support size $K = 20$ (2-way, 10-shot), query size $Q = 16$; meta-batch $B = 9$; epochs $E = 2000$.
Output: shared encoder and head parameters ϕ .

- 1: Initialise ϕ : GIN weights either random or from the pre-trained checkpoint, head weights $\{a_{jm}, b_{jm}, w_{jm}\}$ random
- 2: **for** $e = 1$ to E **do**
- 3: Sample a meta-batch of B assays $T_1, \dots, T_B \sim \mathcal{T}$
- 4: **for** each assay T_t **do**
- 5: Draw a class-balanced support set S_t (K molecules) and a query set Q_t
- 6: $u(x) \leftarrow [h_G(x) \parallel f(x)] \in \mathbb{R}^{2348}$ {GIN read-out $\parallel$ ECFP4}
- 7: $z_i \leftarrow h(u(x_i))$ by Eq. (1) {bounded branch head, $z \in \mathbb{R}^{2048}$ }
- 8: $\ell_0 \leftarrow (\frac{1}{2} \text{median}_{i < j} \|z_i - z_j\|^2)^{1/2}$ {median heuristic}
- 9: $\psi_t^* \leftarrow \arg \max_{\psi} J_t(\psi \mid \phi)$ on S_t , by L-BFGS from $\psi_0 = (\log \ell_0, 0, \log 0.1)$ {inner fit, Eq. (13)}
- 10: $\mathcal{L}_{\text{val}}^t \leftarrow$ predictive loss of the GP with (ψ_t^*, ϕ) on Q_t {outer evaluation}
- 11: $g_t \leftarrow \nabla_{\phi} \mathcal{L}_{\text{val}}^t$ by Eq. (2) {implicit-function-theorem hyper-gradient; inverts the $3 \times 3 \nabla_{\psi}^2 J_t$ }
- 12: **end for**
- 13: $\phi \leftarrow \text{AdamW}(\phi, \frac{1}{B} \sum_t g_t)$, gradients clipped to norm 1.0
- 14: **if** $e \bmod 10 = 0$ **then**
- 15: Evaluate on the held-out meta-test assays
- 16: **end if**
- 17: **end for**
- 18: **return** ϕ

entry of that matrix stays away from zero, and the two ablation arms that abort in Table 2 abort inside line 11.

B The Hypergradient

Equation (2) is an application of the implicit function theorem, which we state and derive here for completeness; the implementation follows [6].

B.1 The implicit function theorem

Let $F(x, y)$ be a C^1 function, $F : U \rightarrow \mathbb{R}^m$, defined on an open set U of $\mathbb{R}^{n+m}$. If $F(x_0, y_0) = 0$ and the partial derivative $\frac{\partial F}{\partial y}(x_0, y_0)$ with respect to y is invertible, then there is a neighbourhood V of x_0 on which y can be written as a function of x , $y = \varphi(x)$, with $F(x, \varphi(x)) = 0$.

Step 1. Take a first-order Taylor expansion of $F(x, y)$ at (x_0, y_0) :

$$F(x, y) \approx F(x_0, y_0) + D_x F \Delta x + D_y F \Delta y,$$

where $\Delta x = x - x_0$ and $\Delta y = y - y_0$.

Step 2. Set $F(x, y) = 0$ and solve for y . Since $F(x_0, y_0) = 0$,

$$0 \approx D_x F(x_0, y_0)(x - x_0) + D_y F(x_0, y_0)(y - y_0),$$

so if $D_y F(x_0, y_0)$ is invertible,

$$y \approx y_0 - [D_y F(x_0, y_0)]^{-1} D_x F(x_0, y_0)(x - x_0).$$

Step 3. Define the mapping $G(x, y) = y - [D_y F(x_0, y_0)]^{-1} F(x, y)$, which is a contraction on a neighbourhood of (x_0, y_0) . By the Banach fixed-point theorem it has a unique fixed point $y = \varphi(x)$ there, and $F(x, \varphi(x)) = 0$.

Step 4. Differentiate $F(x, \varphi(x)) = 0$ with respect to x :

$$\frac{d}{dx} F(x, \varphi(x)) = D_x F(x, \varphi(x)) + D_y F(x, \varphi(x)) \cdot \frac{d\varphi}{dx} = 0,$$

whence

$$\frac{d\varphi}{dx} = -[D_y F(x, \varphi(x))]^{-1} D_x F(x, \varphi(x)). \quad (7)$$

B.2 Application to the inner fit

In our setting the roles are the reverse of the usual reading, and it is worth stating them explicitly because the direction is easy to get backwards. The *inner* variables are the three kernel hyperparameters ψ , which are refitted per assay and not shared; the *outer* variables are the encoder and head parameters ϕ , which are shared across assays and are what meta-training returns. The inner solution $\psi_t^*(\phi) = \arg \max_{\psi} J_t(\psi | \phi)$ therefore depends implicitly on ϕ .

Applying (7) to the stationarity condition $F(\phi, \psi) = \nabla_{\psi} J_t(\psi | \phi) = 0$ at $\psi = \psi_t^*$ gives

$$\frac{d\psi_t^*}{d\phi} = -[\nabla_{\psi}^2 J_t]^{-1} \nabla_{\psi} \nabla_{\phi} J_t, \quad (8)$$

and composing with $\nabla_{\psi} \mathcal{L}_{\text{val}}^t$ by the chain rule recovers Eq. (2). Gradients thus reach ϕ from the outer loss without differentiating through the optimisation trajectory that produced ψ_t^* , which is what makes a 2000-epoch bi-level run affordable at all.

Two remarks. First, (8) is exactly where a 3×3 matrix gets inverted once per task per epoch, and where meta-training aborts with `torch.linalg.solve: the input matrix is singular` when it does. Second, the theorem requires only invertibility of $\nabla_{\psi}^2 J_t$ at the solution — it says nothing about how the feature map should be built to keep it invertible. Section 4 is our attempt to say something about that, and Appendix G reports how far the attempt gets.

C The MoleculeNet Benchmark

MoleculeNet [28] is a benchmark suite and evaluation platform for molecular machine learning. It collects public molecular datasets spanning bioactivity prediction, physicochemical property estimation and toxicity assessment, and supplies for each of them a standardised processing pipeline, featurisation and evaluation metric, so that different modelling algorithms can be compared fairly and reproducibly. We use four of its panels, each of which is a set of binary assays over a shared compound library. Sizes and the meta-train / meta-test assay split are given in Table 4; the split itself is the one introduced by PAR [27] and adopted by ADKF-IFT [6],

Table 4: MoleculeNet panels used, with the meta-train/meta-test assay split. ToxCast’s usable split is 362/156 rather than 450/167 after the shared preprocessing filters 99 assays.

	Tox21	SIDER	MUV	ToxCast
Compounds	8,014	1,427	93,127	8,615
Assays	12	27	17	617
Meta-train assays	9	21	12	362
Meta-test assays	3	6	5	156

which is what makes our numbers comparable to the published ones.

Tox21 collects nuclear-receptor and stress-response toxicity assays over 8,014 compounds. It is the most evenly sized of the four in both dimensions, which is why we read the ablations primarily off it. **SIDER** records side effects of marketed drugs grouped by system organ class; at 1,427 compounds it is by far the smallest, and its meta-test assays are the most heterogeneous — Figure 10 shows them spanning more than ten AUROC points. **MUV** is the largest at 93,127 compounds, and was constructed from PubChem specifically to be unbiased for virtual-screening validation, which leaves each assay with very few actives (roughly 29); Appendix L explains why that makes the panel uninformative under this protocol. **ToxCast** has by far the most assays — 617 in the raw release — from in-vitro high-throughput toxicology.

One difference from the numbers usually quoted alongside this split deserves to be stated plainly. The ToxCast division is normally reported as 450 meta-training and 167 meta-testing assays. The shared preprocessing code filters out 99 assays before any model sees them, so the split that is actually run is 362/156, and that is what Table 4 reports. We use the usable figure rather than the nominal one because it is the one our runs correspond to.

D Evaluation Metric

On the MoleculeNet benchmark, **AUROC** (area under the receiver operating characteristic curve) is the primary metric for molecular classification tasks. AUROC measures a model’s ability to separate positive from negative samples across all classification thresholds: it is the probability that a randomly chosen positive is ranked above a randomly chosen negative. Its value lies in $[0, 1]$, higher is better, and 0.5 corresponds to random guessing. Because many molecular property prediction tasks are severely class-imbalanced, AUROC gives a threshold-independent assessment of overall performance, which is why it is the standard choice in this benchmark’s evaluation protocol. Given n_{pos} positive samples, n_{neg} negative samples, and model scores s ranking all samples,

$$\text{AUROC} = \frac{1}{n_{\text{pos}} \times n_{\text{neg}}} \sum_{i=1}^{n_{\text{pos}}} \sum_{j=1}^{n_{\text{neg}}} \mathbb{I}(s_i > s_j), \quad (9)$$

where s_i and s_j are the scores assigned to positive and negative samples respectively and $\mathbb{I}(\cdot)$ is the indicator function, equal to 1 if $s_i > s_j$ and 0 otherwise.

We report AUROC in percent, averaged over the meta-test assays of a panel. Two conventions on top of that are inherited from the benchmark and are worth restating here, since both inflate the

headline figure: the reported number is the *best* such average over the evaluation rounds of a run rather than the last one (Appendix M gives both for every cell), and the per-task adaptation step at evaluation time has access to labels drawn from the same pool as the molecules being scored (Appendix L quantifies what that is worth).

E Baselines

Baseline numbers in Table 1 are quoted from the original publications as collected in [6]; we did not re-run them. They fall into three groups, following the grouping used in Section 5.1.

Metric-based few-shot learners. These learn a feature space in which distance between two molecules reflects how alike they are for the endpoint at hand, and classify a query by its position relative to the support set. **Siamese** networks [14] compare pairs of examples through shared encoders and a fixed similarity function. **ProtoNet** [23] forms a centroid per class and assigns a query to the nearest one. **TPN** [18] builds a graph over support and query points together and propagates labels along it, making the inference transductive. **EGNN** [13] recasts few-shot classification as edge-label prediction on a relational graph, which lets intra- and inter-class edges be learned explicitly. **PAR** [27] is the molecular specialisation of this family: it learns property-aware molecular embeddings and an adaptive relation graph over the support set. Our method sits closest to this group in spirit — a Gaussian process with a stationary kernel is a distance-based learner — which is why the geometry of the space that distance is taken in is the object of this paper.

Optimisation-based meta-learners. These learn something that adapts quickly to a new task under a small number of gradient steps. **MAML** [7] learns an initialisation from which a few steps of gradient descent suffice on an unseen task. **IterRefLSTM** [3] refines support and query embeddings jointly through an iterative LSTM so that the two co-adapt before a one-shot prediction is made. **ADKF-IFT** [6] adapts only the kernel hyper-parameters of a deep-kernel Gaussian process per task and propagates gradients to the shared encoder with the implicit function theorem; it is the framework this paper builds on, and Algorithm 1 differs from it in one line. **UniMatch** [15] adds hierarchical matching over atoms and over tasks on top of a gradient-based adaptation scheme.

Graph-based and hybrid approaches. These bring additional structural or contextual information to bear. **MixHop** [1] mixes multiple neighbourhood powers within a single graph convolution, so a layer can represent higher-order interactions that repeated first-order aggregation cannot. **CHEF** [2] fuses representations taken at several depths of the network through a Hebbian ensemble, letting whichever level of abstraction suits the new domain dominate; it is included as a cross-domain few-shot baseline. **MHNfs** [22] enriches each molecule’s representation by retrieving a large context set of reference molecules with a modern Hopfield network, that is, by associative recall over an external library rather than by additional message passing.

Pre-trained protocol. The lower block of Table 1 initialises the encoder from the supervised context-prediction checkpoint

of [12], reported there as **Pre-GNN**. **GNN-MAML** [10] and **Pre-PAR** [27] are the pre-trained counterparts of the corresponding entries above, and **MolFeSCue** [30] adds a contrastive objective with a dynamic focal-style loss aimed specifically at the imbalanced, data-limited regime. We report both protocols because the pre-trained initialisation changes the ordering of several methods, and because our own ablations are run from scratch, where the encoder cannot be credited for the result.

F Proofs

F.1 Setup and notation

A task supplies a support set $\{(x_i, y_i)\}_{i=1}^n$ with $n = 20$ (2-way, 10-shot). Each molecule is encoded to $\phi(x_i) \in \mathbb{R}^p$, $p = 2348$, by concatenating the 300-dimensional GIN read-out with the 2048-bit ECFP4 fingerprint, and is then passed through the head $h : \mathbb{R}^p \rightarrow \mathbb{R}^d$, $d = 2048$,

$$h_j(x) = \sum_{m=1}^M A_{jm} \zeta(a_{jm}^\top \phi(x) + b_{jm}), \quad A_{jm} = \tanh(w_{jm}), \quad (10)$$

for $j = 1, \dots, d$, with $\zeta(t) = (1 + e^{-t})^{-1}$ the logistic function and $M = 30$. We abbreviate $z_i = h(x_i)$ and write $A_j \in \mathbb{R}^M$ for the j -th row of the gate matrix.

The Gaussian process has zero mean and covariance

$$\Sigma = K + \sigma_n^2 I, \quad K_{ij} = s k(r_{ij}), \quad r_{ij} = \frac{\|z_i - z_j\|_2}{\ell}, \quad (11)$$

where k is the Matérn kernel with smoothness $\nu = 5/2$,

$$k(r) = \left(1 + \sqrt{5}r + \frac{5}{3}r^2\right)e^{-\sqrt{5}r}, \quad k(0) = 1, \quad 0 < k \leq 1, \quad (12)$$

$s > 0$ is the output scale and $\ell > 0$ the (single, shared) length-scale. This matches the implementation exactly: `ScaleKernel(MaternalKernel())` with `ard_num_dims=None` and a `ZeroMean`. The inner objective is the log marginal likelihood

$$J(\psi) = -\frac{1}{2}y^\top \Sigma^{-1}y - \frac{1}{2}\log \det \Sigma - \frac{n}{2}\log 2\pi, \quad (13)$$

with $\psi = (\log \ell, \log s, \log \sigma_n^2)$, maximised over ψ for fixed head parameters. The outer problem differentiates through this optimum using the implicit function theorem, which requires $\nabla_\psi^2 J$ to be invertible.

F.2 Proof of Proposition 4.1

Proof. Fix j and x, x' . Writing $\varsigma_{jm}(x) = \zeta(a_{jm}^\top \phi(x) + b_{jm})$ and using (10),

$$\begin{aligned} |h_j(x) - h_j(x')| &= \left| \sum_{m=1}^M A_{jm} (\varsigma_{jm}(x) - \varsigma_{jm}(x')) \right| \\ &\leq \sum_{m=1}^M |A_{jm}| |\varsigma_{jm}(x) - \varsigma_{jm}(x')|. \end{aligned}$$

Since ς takes values in the open interval $(0, 1)$, every difference $|\varsigma_{jm}(x) - \varsigma_{jm}(x')|$ is bounded by 1, giving $|h_j(x) - h_j(x')| \leq \sum_m |A_{jm}| = \|A_j\|_1$. Squaring and summing over j ,

$$\|h(x) - h(x')\|_2^2 = \sum_{j=1}^d |h_j(x) - h_j(x')|^2 \leq \sum_{j=1}^d \|A_j\|_1^2 = R_\star^2.$$

Finally $|A_{jm}| = |\tanh(w_{jm})| < 1$ gives $\|A_{j\cdot}\|_1 < M$ for every j , hence $R_\star < M\sqrt{d}$. $\square$

Remark F.1 (The constant is loose but the existence is the point). With $M = 30$ and $d = 2048$ the worst case is $M\sqrt{d} \approx 1358$, and the realised value at initialisation is $R_\star \approx 1265$, whereas the diameter actually attained on SIDER support sets is ≈ 9.8 (Table 6). The bound is therefore very conservative. Its role is not to be tight: it is that a finite, data-independent constant *exists at all*, so that no sequence of encoder updates can drive the pairwise distances to infinity. For a linear head $h(x) = V\phi(x)$ the corresponding quantity is $\|V\|_{\text{op}} \text{diam } \phi(X)$, which the outer optimiser can increase without limit.

F.3 Proof of Proposition 4.2

Proof. Differentiating (12),

$$\begin{aligned} k'(r) &= (\sqrt{5} + \frac{10}{3}r)e^{-\sqrt{5}r} - \sqrt{5}\left(1 + \sqrt{5}r + \frac{5}{3}r^2\right)e^{-\sqrt{5}r} \\ &= -\frac{5}{3}r(1 + \sqrt{5}r)e^{-\sqrt{5}r}, \end{aligned}$$

where the leading terms cancel. Hence $g(r) = -rk'(r) = \frac{5}{3}r^2(1 + \sqrt{5}r)e^{-\sqrt{5}r} \geq 0$, with $g(r) = 0$ iff $r = 0$.

Substitute $u = \sqrt{5}r$, so that $g = \frac{1}{3}u^2(1 + u)e^{-u} =: \tilde{g}(u)$. Then

$$\tilde{g}'(u) = \frac{1}{3}e^{-u}\left[(2u + 3u^2) - (u^2 + u^3)\right] = \frac{1}{3}e^{-u}u(2 + 2u - u^2).$$

For $u > 0$ the sign of $\tilde{g}'$ is that of $2 + 2u - u^2$, a downward parabola with roots $u = 1 \pm \sqrt{3}$. Only $u^\star = 1 + \sqrt{3}$ is positive, so $\tilde{g}$ increases on $(0, u^\star)$ and decreases on $(u^\star, \infty)$: $\tilde{g}$, and therefore g , is unimodal with the unique maximiser

$$r^\star = \frac{u^\star}{\sqrt{5}} = \frac{1 + \sqrt{3}}{\sqrt{5}} \approx 1.221810.$$

Its value is $g(r^\star) = \frac{1}{3}(1 + \sqrt{3})^2(2 + \sqrt{3})e^{-(1+\sqrt{3})} \approx 0.604351$. The asymptotics follow by inspection: $g(r) = \frac{5}{3}r^2 + O(r^3)$ as $r \rightarrow 0$, and $g(r) = \frac{5\sqrt{5}}{3}r^3e^{-\sqrt{5}r}(1 + O(r^{-1}))$ as $r \rightarrow \infty$. $\square$

F.4 Proof of Theorem 4.3

We first record two standard facts.

Lemma F.2 (Fisher information of a zero-mean Gaussian). *For $y \sim \mathcal{N}(0, \Sigma(\psi))$ the Fisher information is $\mathcal{I}_{ab} = \frac{1}{2} \text{tr}(\Sigma^{-1}\Sigma_a\Sigma^{-1}\Sigma_b)$ with $\Sigma_a = \partial_a\Sigma$. In particular $\mathcal{I}_{\ell\ell} = \frac{1}{2} \text{tr}[(\Sigma^{-1}\partial_{\log \ell}K)^2]$, and $\mathcal{I}_{\ell\ell}$ equals the expectation of $-\partial_{\log \ell}^2 J$.*

Lemma F.3 (Spectral bound on the support covariance). $\lambda_{\max}(\Sigma) \leq ns + \sigma_n^2$ and $\lambda_{\min}(\Sigma) \geq \sigma_n^2$.

Proof. K has constant diagonal $K_{ii} = sk(0) = s$ and, because $0 < k \leq 1$, off-diagonal row sums at most $(n-1)s$. Gershgorin's theorem places every eigenvalue of K in $[s - (n-1)s, s + (n-1)s]$, so $\lambda_{\max}(K) \leq ns$. Adding the nugget gives the upper bound. For the lower bound, K is a Gram matrix of a positive-definite kernel, hence $K \succeq 0$ and $\lambda_{\min}(\Sigma) \geq \sigma_n^2$. $\square$

Proof of Theorem 4.3. From (11) and the chain rule, $\partial_{\log \ell} K_{ij} = sk'(r_{ij}) \cdot \partial_{\log \ell} r_{ij} = -sr_{ij}k'(r_{ij}) = sg(r_{ij})$, since $r_{ij} \propto \ell^{-1}$ gives

$\partial_{\log \ell} r_{ij} = -r_{ij}$. Because $g(0) = 0$ the diagonal of $\partial_{\log \ell} K$ vanishes, so

$$\|\partial_{\log \ell} K\|_F^2 = s^2 \sum_{i \neq j} g(r_{ij})^2. \quad (14)$$

Let $B = \Sigma^{-1/2}(\partial_{\log \ell} K)\Sigma^{-1/2}$, which is symmetric because $\Sigma \succ 0$. Then $\text{tr}[(\Sigma^{-1}\partial_{\log \ell} K)^2] = \text{tr}(B^2) = \|B\|_F^2$, and since $\|AXA\|_F \geq \sigma_{\min}(A)^2\|X\|_F$ with $\sigma_{\min}(\Sigma^{-1/2}) = \lambda_{\max}(\Sigma)^{-1/2}$,

$$\mathcal{I}_{\ell\ell} = \frac{1}{2}\|B\|_F^2 \geq \frac{\|\partial_{\log \ell} K\|_F^2}{2\lambda_{\max}(\Sigma)^2}. \quad (15)$$

By hypothesis $\|z_i - z_j\| \geq q$ for $i \neq j$, and by Proposition 4.1 $\|z_i - z_j\| \leq R_\star$; dividing by ℓ gives $r_{ij} \in [q/\ell, R_\star/\ell]$ for all $i \neq j$. Proposition 4.2 states that g is unimodal, so on any interval its minimum is attained at an endpoint:

$$g(r_{ij}) \geq \underline{g} := \min\{g(q/\ell), g(R_\star/\ell)\} > 0.$$

Substituting into (14) and using that there are $n(n-1)$ off-diagonal entries, $\|\partial_{\log \ell} K\|_F^2 \geq s^2 n(n-1) \underline{g}^2$. Combining with (15) and Lemma F.3 yields

$$\mathcal{I}_{\ell\ell} \geq \frac{n(n-1)s^2 \underline{g}^2}{2(ns + \sigma_n^2)^2} > 0.$$

Finally, $\mathcal{I}_{\ell\ell}$ is the $(\log \ell, \log \ell)$ entry of the expected negative Hessian $-\mathbb{E}\nabla_{\psi}^2 J$; a vanishing diagonal entry of a positive semi-definite matrix forces the corresponding row and column to vanish, so $\mathcal{I}_{\ell\ell} > 0$ is necessary for $-\mathbb{E}\nabla_{\psi}^2 J$ to be non-singular. $\square$

Remark F.4 (Scope). Theorem 4.3 bounds one diagonal entry away from zero, which is necessary but not sufficient for the full 3×3 Hessian over $(\log \ell, \log s, \log \sigma_n^2)$ to be invertible. We claim only what is proved. Empirically, the failure mode we observe in practice is precisely a degenerate length-scale direction: the arms that abort do so inside the implicit-function-theorem solve, raising `torch.linalg.solve`: the input matrix is singular, whereas the proposed head does not. Which arms those are does not follow boundedness — the bounded single-branch head aborts and the unbounded branched head completes — so we do not claim the theorem explains the pattern (Section 5.1, Appendix G).

F.5 Why we do not state a condition-number theorem

It is tempting to advertise a bound on $\kappa(\Sigma)$. Lemma F.3 immediately gives

$$\kappa(\Sigma) = \frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Sigma)} \leq \frac{ns + \sigma_n^2}{\sigma_n^2}, \quad (16)$$

which for $n = 20$ and the implementation's $\sigma_n^2 = 0.1$ is a finite constant. We deliberately do *not* present (16) as a contribution: it uses only positive semi-definiteness of K and the nugget, so it holds for *every* feature map, bounded or not, including maps that destroy all information. A bound that a degenerate architecture also satisfies cannot distinguish architectures. The quantity that does discriminate is the length-scale information (6), which is why the argument is routed through it.

Table 5: Matérn-5/2 length-scale information profile $g(r) = \frac{5}{3}r^2(1 + \sqrt{5}r)e^{-\sqrt{5}r}$, normalised by its maximum $g(r^*) = 0.60435$ at $r^* = 1.22181$.

r	0.01	0.1	0.5	1.0	r^*	$\sqrt{2}$	2.0	5.0
$g(r)$	1.67×10^{-4}	0.0163	0.2885	0.5764	0.6044	0.5873	0.4167	0.0071
$g(r)/g(r^*)$	0.0003	0.027	0.477	0.954	1.000	0.972	0.689	0.012

F.6 Sharpness of Remark 4.4

Proposition F.5 (Boundedness is not sufficient). *There exists a head satisfying Proposition 4.1 for which $\mathcal{I}_{\ell\ell} = 0$ and rank $K = 1$.*

Proof. Take $a_{jm} = 0$ for all j, m . Then $\varsigma_{jm}(x) = \varsigma(b_{jm})$ does not depend on x , so $h_j(x) = \sum_m A_{jm}\varsigma(b_{jm})$ is a constant and $h \equiv z_0$ for every input. Proposition 4.1 holds (the realised diameter is 0). But then $r_{ij} = 0$ for all i, j , so $K = s\mathbf{1}\mathbf{1}^\top$, which has rank 1, and $\partial_{\log \ell} K_{ij} = s g(0) = 0$ identically, so $\mathcal{I}_{\ell\ell} = \frac{1}{2} \text{tr}[(\Sigma^{-1} \cdot 0)^2] = 0$. The length-scale is then completely unidentifiable: J does not depend on ℓ at all. $\square$

This is why Theorem 4.3 carries the separation hypothesis, and why we do not claim the architecture guarantees well-posedness on its own. Boundedness closes the $r \rightarrow \infty$ failure mode structurally; the $r \rightarrow 0$ mode is left to be checked. Section G reports the realised separation.

F.7 Proof of Corollary 4.5

Proof. The implementation sets $\ell_0 = (\frac{1}{2} \text{median}_{i < j} \|z_i - z_j\|^2)^{1/2}$. Writing $D = \text{median}_{i < j} \|z_i - z_j\|$ and noting that the median commutes with the increasing map $t \mapsto t^2$, we get $\ell_0 = D/\sqrt{2}$ and hence a median normalised distance $r_{\text{med}} = D/\ell_0 = \sqrt{2}$, independent of the data.

By Proposition 4.2, g attains its maximum at $r^* = (1 + \sqrt{3})/\sqrt{5}$. Using $\tilde{g}(u) = \frac{1}{3}u^2(1 + u)e^{-u}$ with $u = \sqrt{5}r$, the median pair has $u = \sqrt{10}$ and

$$\frac{g(\sqrt{2})}{g(r^*)} = \frac{10(1 + \sqrt{10})e^{-\sqrt{10}}}{(1 + \sqrt{3})^2(2 + \sqrt{3})e^{-(1+\sqrt{3})}} = 0.97176 \dots$$

so the median support pair sits at 97.2% of the peak of g , and at 94.4% of the peak Fisher information $\mathcal{I}_{\ell\ell} \propto g^2$. $\square$

Table 5 lists the profile at several radii; the plateau around r^* is broad, which is why a heuristic that lands at $\sqrt{2}$ rather than exactly at r^* loses so little.

G Realised Geometry of the Support Sets

Theorem 4.3 requires a separation $q > 0$ that the architecture does not supply, so we measure it. Table 6 reports, for each head, the diameter of the mapped support set and the range of normalised distances $r_{ij} = \|z_i - z_j\|/\ell_0$ after the median initialisation, averaged over the six SIDER meta-test tasks.

Two observations, both of which we report as they stand.

First, the median initialisation is doing a great deal of work: because ℓ_0 is set from the data, the *absolute* scale of the feature map is divided out, and every head begins meta-training with its typical pair near $r = \sqrt{2}$ and mean information within 94–97% of

the maximum. At initialisation the heads are therefore comparably conditioned, and the theory above does not separate them.

Second, the separation that matters is the one maintained *during* meta-training, once the encoder and the head are being updated by the outer loop. This is where the heads diverge: the two that abort do so after several hundred epochs, not at the start. We therefore present Proposition 4.1 as a statement about what the optimiser is prevented from doing over the course of training, and Table 6 as evidence that the initial conditions are matched — not as evidence for the mechanism itself, which would require tracking $\mathcal{I}_{\ell\ell}$ along the whole trajectory.

H Full Experimental Configuration

This section carries the detail removed from Section 5.1: what a reader needs in order to reproduce the work rather than to read the results.

Encoder and head. The encoder is a five-layer GIN with 300-dimensional embeddings, last-layer readout, mean pooling, batch normalisation and dropout 0.5; under the pre-trained protocol it is initialised from the supervised context-prediction checkpoint of [12]. Fingerprints are 2048-bit ECFP4 (Morgan, radius 2). The head uses $M = 30$ branches and outputs $d = 2048$ dimensions.

Gaussian process and optimisation. The Gaussian process is exact, with a zero mean and a Matérn-5/2 kernel carrying a single shared length-scale, an output scale and Gaussian observation noise. The noise parameter has a log-normal prior with mode 0.1, and the length-scale is initialised by the median heuristic. The inner loop maximises the log marginal likelihood over the three kernel hyper-parameters with L-BFGS. The outer loop uses AdamW at learning rate 10^{-3} and weight decay 5×10^{-5} , with gradients clipped to norm 1.0, a meta-batch of 9 assays, one inner and one outer step per episode, and 2000 meta-training epochs evaluated every 10. Table 3 lists the complete configuration.

Compute and model size. The model has 146.2M parameters: the head accounts for 144.4M (98.7%), the GIN encoder for 1.9M, and the Gaussian process for three scalars. We state this plainly because the head is large, and any comparison against a smaller-headed baseline confounds architecture with capacity — which is why the MLP control is matched on parameter count rather than on width. Runs were executed on RTX 3090 and RTX 4060 Ti GPUs. A single run takes 5.6 s per epoch in isolation and 8.5 s per epoch when four runs share two GPUs, so one 2000-epoch SIDER run costs roughly 4.7 hours under the concurrency we used.

The ablation arms in full. The branched unbounded head keeps the fan-out and the summation of the proposed head but drops the squashing. The bounded unbranched head is $\tanh(Wx)$, a single branch. The ℓ_2 -normalised arm is a linear map followed by normalisation to the unit sphere. The parameter-matched MLP is two layers whose hidden width is solved so that its parameter count matches the proposed head's to within one percent. The encoder, the fused input, the Gaussian process and the meta-learning procedure are identical across all five.

Table 6: Geometry of the mapped support sets at initialisation (SIDER, six tasks, pre-trained encoder). R_* is the structural bound of Proposition 4.1; “—” marks heads for which no such architectural constant exists. The last column records whether meta-training completed without the implicit-function-theorem solve becoming singular.

Head	bounded	realised diam.	R_*	$r_{\min} - r_{\max}$	training completed
BOND (bounded, branched)	yes	9.84	1264.5	0.86 – 1.92	yes
tanh, 1 branch	yes	7.65	—	0.64 – 1.99	no
ℓ_2 -norm, 1 branch	yes	1.42	—	0.94 – 1.54	yes
branched, unbounded	no	23.63	—	0.88 – 2.01	yes
MLP, matched	no	2.71	—	0.80 – 1.99	no

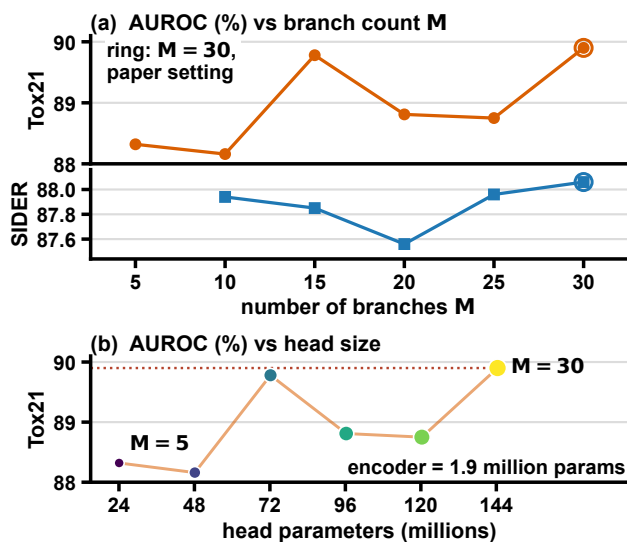


Figure 7: Effect of the number of branches M on predictive performance and parameter count. The sweep comes from an earlier single-seed campaign and is not on the same footing as Table 1; its $M = 30$ points are not the runs reported there.

I Sensitivity to the Number of Branches

Figure 7 varies M , which determines most of the model capacity, and plots performance against parameter count so that the capacity cost is explicit. Performance is not monotonic in M on either dataset. This sweep predates the grid reported in the body: it was run at a single seed, and its $M = 30$ points are not the runs Table 1 reports — its SIDER value sits 1.50 AUROC below the SIDER figure in that table for the same configuration. It is included because the shape of the curve is the point, and that shape does not depend on which campaign produced it; the levels should not be read against Table 1.

J Additional Views of the Main Results

The three figures below present information that also appears in Tables 1 and 2, or that concerns individual assays rather than the comparisons the body draws. They are collected here so that the main text carries only the measurements its argument rests on.

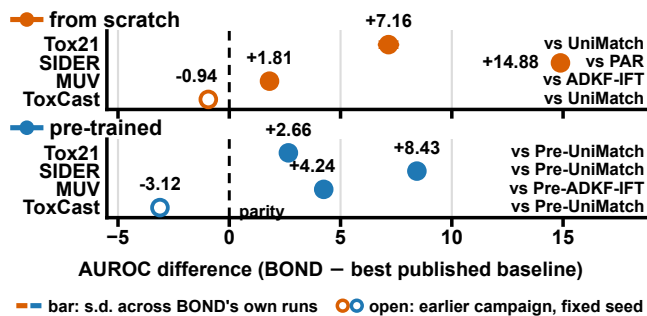


Figure 8: Gain of BOND over the strongest published baseline for each dataset and training protocol. Each interval represents the run-to-run variation of BOND rather than an interval for the difference between means.

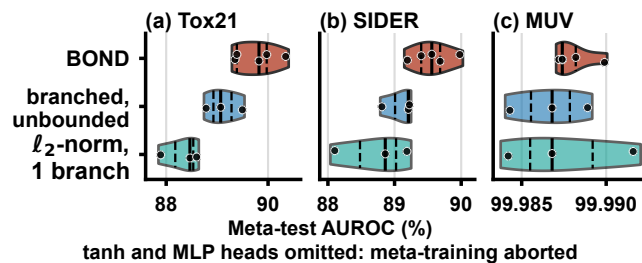


Figure 9: Distribution of the reported metric across feature maps. Violins show the pooled per-seed distributions, horizontal dashes indicate quartiles, and dots represent individual seeds.

Distribution across seeds and feature maps. Figure 9 shows the distributions underlying the means in Table 2, allowing the feature maps to be compared beyond their point estimates.

Per-assay behaviour. Figure 10 decomposes each panel’s headline number into its meta-test assays. The radial axis begins at chance, and every petal is read at the evaluation round the run itself reports, so the petals of a panel average exactly to the value in Table 1.

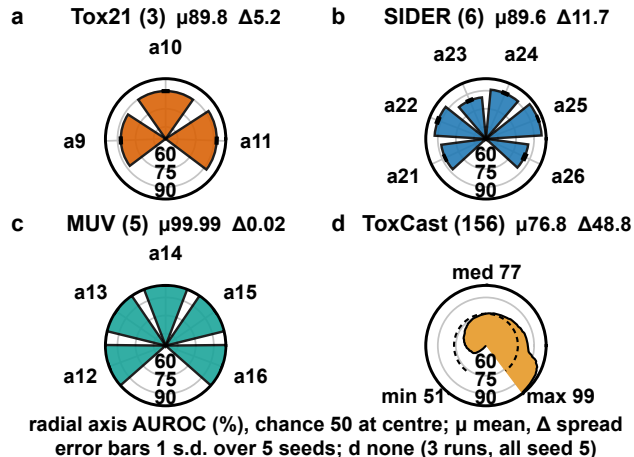


Figure 10: Per-assay AUROC on the meta-test split. Each petal represents one assay, and the radial axis begins at chance performance.

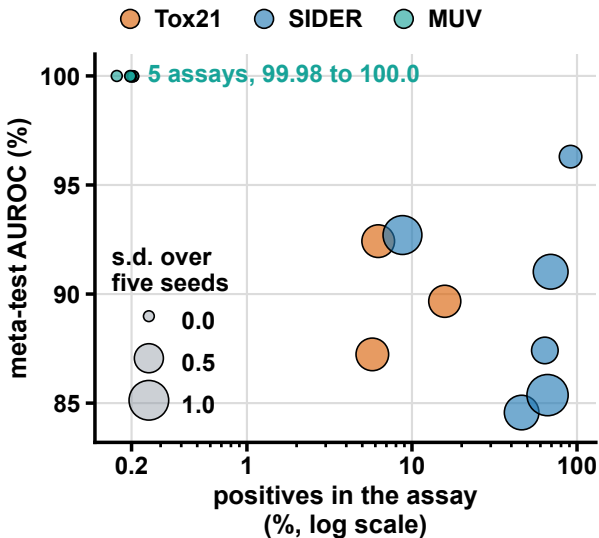


Figure 11: Meta-test assay difficulty against class imbalance, one bubble per assay; area is the spread across runs.

K Assay Difficulty and Class Imbalance

The panel means in Table 1 average over assays that are not measuring comparable problems. Figure 11 makes that checkable: each meta-test assay is one bubble, placed by how imbalanced it is and by the AUROC reached on it, with area giving the seed-to-seed spread. MUV’s assays sit at roughly two positives per thousand molecules and score at the ceiling with a spread under a hundredth of a point; SIDER’s are near-balanced, score ten to fifteen points lower and are two orders of magnitude less stable. A single mean per panel conceals that difference, and it is part of why we read the ablations off Tox21 and SIDER rather than off the average.

Table 7: MUV controls, each row changing one thing. The statistic is the best mean meta-test AUROC over evaluation rounds, as in the main tables.

What changed	Reported AUROC
A released configuration	99.97
B adaptation labels randomly permuted	79.79 ± 1.74
C test-time adaptation disabled	72.13 ± 3.44
PAR, as published [27]	66.48
ECFP4 Tanimoto 1-NN, no learning	68.08

L What Produces the MUV Column

MUV reaches essentially 100 under this protocol for every feature map we tried, so we ran single-variable controls to find out what carries it. All arms are MUV, pre-trained, seed 0, and differ from the released configuration in exactly one respect.

The per-task adaptation step used at evaluation time takes gradient steps on molecules drawn from the same pool as the molecules being scored, and it uses their labels. Row B holds the molecules, the batches, the number of optimiser steps and the gradient clipping fixed and destroys only the label information; the reported figure falls by more than 20 points. Row C removes the adaptation entirely and lands at 72.13, close to a published metric-based baseline and to a learning-free fingerprint nearest-neighbour, neither of which can exploit the channel.

Two qualifications matter. First, this is a property of the evaluation path, which we inherit unmodified from the reference implementation of this benchmark so that our numbers remain comparable to the published ones; we do not claim it accounts for any particular published result, which would require running each of them. Second, the effect is strongly panel-dependent: the same control on SIDER moves the reported figure by roughly 3 AUROC rather than 28. The magnitude tracks how few positives a task has, and MUV is the extreme case at roughly 29 per task. This is why we read the ablations off Tox21 and SIDER and treat MUV as uninformative.

M Both Reported Statistics

Table 8 gives, for every cell, the best-over-evaluations figure quoted in the main table alongside the final-round figure, so that the effect of the benchmark’s reporting convention is visible rather than implicit. The gap is between +0.01 and +1.91 AUROC: one to two points on every unsaturated panel, and essentially nothing on MUV, where the metric is at the ceiling.

Figure 12 shows the same comparison as an agreement analysis. The bias is +1.04 AUROC with limits of agreement [−1.10, +3.19], so the convention is not a wash: it is a systematic offset. It is not, however, a gradient in the mean: pooled over all four panels the difference does fall as the mean rises (Spearman −0.73), but that is entirely MUV, which sits at 99.98 where there is nothing left to select into. Across the three unsaturated panels there is no gradient at all (Pearson −0.04, $p = 0.85$). Selection buys one to two points wherever the metric is not saturated, and nothing where it is.

Table 8: The best-over-rounds statistic this benchmark reports, beside the final-round value, for every cell. The gap is what the reporting convention costs.

Dataset	Protocol	Best	Final	Gap
Tox21	from scratch	89.78 ± 0.37	87.86 ± 0.95	+1.91
Tox21	pre-trained	89.01 ± 0.23	87.11 ± 1.28	+1.89
SIDER	from scratch	89.56 ± 0.26	88.37 ± 1.21	+1.19
SIDER	pre-trained	88.77 ± 0.30	88.04 ± 1.05	+0.74
MUV	from scratch	99.99 ± 0.00	99.97 ± 0.01	+0.01
MUV	pre-trained	99.98 ± 0.00	99.98 ± 0.01	+0.01
ToxCast	from scratch	76.80 ± 0.12	76.04 ± 0.19	+0.76
ToxCast	pre-trained	78.51 ± 0.01	76.78 ± 0.04	+1.73

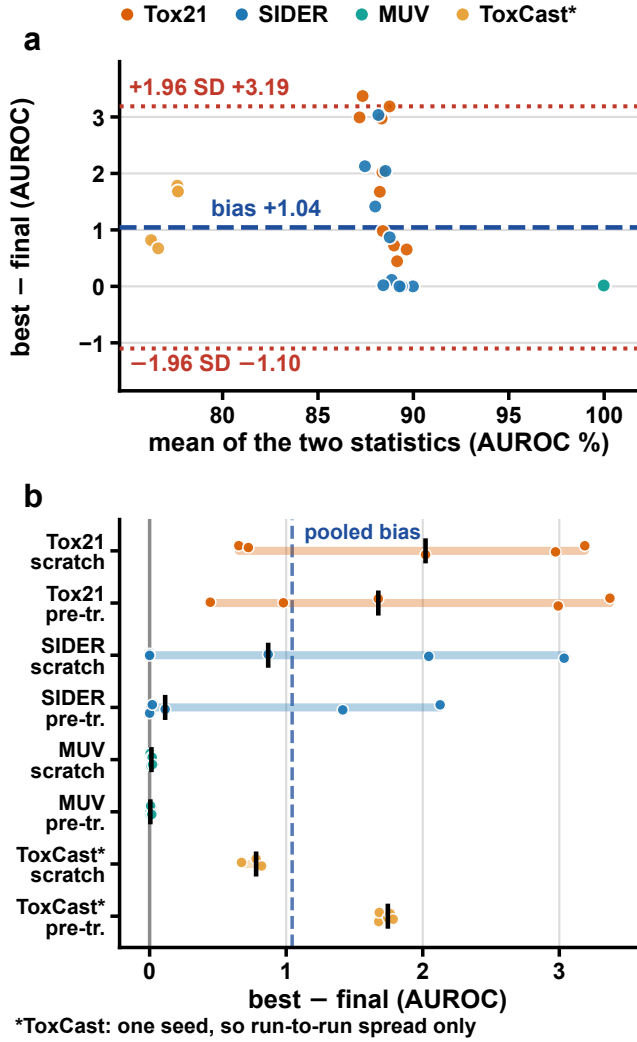


Figure 12: Agreement between the best-over-rounds and final-round statistics over every complete run. (a) Bland–Altman, with the bias and 95% limits of agreement. (b) The same differences per cell.

N Limitations and Future Work

Section 6.1 states the constraints on these results in short form. This section gives the measurement behind each and, where we can, says what experiment would settle it. The order runs from the ones that cut hardest against our own framing to the ones that merely bound the scope.

The median heuristic absorbs most of what the bound constrains. This is the most consequential of the lot. The length-scale is initialised as $\ell_0 = (\frac{1}{2} \text{median}_{i < j} \|z_i - z_j\|^2)^{1/2}$, which is exactly scale-equivariant: multiply the feature map by any constant and ℓ_0 scales with it, leaving every r_{ij} — and therefore the whole kernel matrix — unchanged. Proposition 4.1 bounds absolute diameter, so the heuristic divides out precisely the quantity the proposition controls. Figure 6b measures the consequence: after normalising each pairwise distance by its own median, three heads whose raw diameters differ by four orders of magnitude have coefficients of variation of 0.30, 0.30 and 0.29, having started at 0.16, 0.16 and 0.05. What reaches the kernel is close to identical, and the three accuracies land within one AUROC point of each other. The bound is therefore not idle — it still rules out the $r \rightarrow \infty$ regime over the course of training, and the median initialisation only acts at the start of each inner fit — but on this benchmark the heuristic is doing more of the stabilising work than the architecture is, and we would rather say so than let the table ordering imply otherwise. The experiment that would separate them is to hold ℓ fixed at a common constant across heads, removing the equivariance, and re-run the ablation; we have not run it.

The fingerprint dominates the output distance, and the head does not fix that. Section 1 motivates the architecture partly by the incommensurability of the two concatenated views. Splitting the squared pairwise distance at the head’s input gives the 2048-bit fingerprint 83–92% of it on all four panels, as expected. But the same measurement taken *after* the head, by zeroing one block and re-running, leaves the output distance correlating 0.96 with the fingerprint alone and 0.10 with the graph alone on SIDER. The per-coordinate joint parameterisation is thus the mechanism by which both views reach the kernel at all, not a rebalancing of their scales, and we do not claim the latter. The ablation agrees: removing the fingerprint costs 16.3, 30.7 and 12.7 AUROC on Tox21, SIDER and MUV, an order of magnitude more than any substitution of the head. A fusion that genuinely equalised the two blocks would have to be evaluated by that block-influence statistic rather than by accuracy, since accuracy here mostly reports fingerprint quality.

The arms that abort are the arms we cannot measure. Figure 5a tracks the diameter of the mapped support set for the three heads that complete meta-training. The two that abort — the parameter-matched MLP at epoch 70–340 and the single-branch tanh head at epoch 1420–1920 — have no geometric measurement at all, because the runs that would have produced it are the runs that stopped. We therefore cannot attribute those aborts to the mechanism Theorem 4.3 names, only observe that they occur inside the solve the theorem is about. Worse for the framing we began with, the pattern across arms does not follow boundedness: the bounded tanh head aborts and the unbounded branched head completes 9/9. Across these five arms it is the branch structure, not the bound, that

tracks whether the inner solve stays well posed. Settling this means logging $\mathcal{I}_{\ell t}$, or the smallest eigenvalue and null direction of $\nabla_{\psi}^2 J_t$, over the epochs preceding each failure — cheap to instrument, and the first thing we would add. We also note that the ablation grid covers Tox21, SIDER and MUV only; no ablation arm was run to completion on ToxCast, so nothing in Table 2 speaks to that panel.

MUV carries no information about any of this. Every feature map we tried reaches 99.99 on MUV, including the controls the theory predicts should degenerate, so the column cannot separate arms. Appendix L isolates the cause: withholding label information from the per-task adaptation step at evaluation drops the reported figure from 99.97 to 79.79, and removing the adaptation entirely drops it to 72.13 — close to a published metric-based baseline at 66.48 and to a learning-free ECFP4 nearest neighbour at 68.08. The effect is strongly panel-dependent, worth roughly 28 AUROC on MUV and roughly 3 on SIDER, and its magnitude tracks how few positives an assay has. There is a second, independent reason to distrust the panel: our own head contracts the mapped support set to a diameter of 1.1×10^{-4} on MUV — further into the degenerate $r \rightarrow 0$ regime than the ℓ_2 -normalised head it is meant to improve on — and still scores 99.99. Geometry and accuracy are simply decoupled there. We keep the column for comparability with the published baselines and draw no conclusion from it. The clean version of this experiment reports MUV under an evaluation path that withholds meta-test labels from adaptation, alongside the inherited path so that both are on record; we left the path unmodified so our numbers stay comparable, which means we did not run it.

No fingerprint-only control was run. We ablate the fingerprint but not the encoder. Given that the fingerprint carries 83–92% of the input distance and 0.96 of the output correlation, the obvious control is a fingerprint-only arm at matched capacity, and its absence leaves open the possibility that the graph encoder contributes little under this head. The cost is the reason it is missing rather than any principle: a single 2000-epoch SIDER run takes about 4.7 hours under the concurrency we used, and the control is only informative across five seeds, two protocols and at least the two panels we read the ablations from.

The theory is narrower than the architecture. Three restrictions. Theorem 4.3 bounds one diagonal entry of the Fisher information, which is necessary but not sufficient for the full 3×3 Hessian to be invertible; the constant $R_{\star} \approx 1265$ exceeds the realised SIDER diameter of ≈ 9.8 by two orders of magnitude, and the guarantee decays as $r^3 e^{-\sqrt{5}r}$, so evaluated at $R_{\star}$ it is numerically negligible and the theorem is best read as identifying which geometric quantity governs identifiability rather than as a usable certificate; and the whole analysis assumes an isotropic stationary kernel with a single shared length-scale. An ARD or deep kernel would change which geometric quantity matters, and we would not expect the argument to transfer unchanged.

Future work. Five things follow, in the order we would do them. (1) Instrument $\mathcal{I}_{\ell t}$ and the spectrum of $\nabla_{\psi}^2 J_t$ along the entire meta-training trajectory of every arm, which converts the observed association between head structure and abort into a mechanism or refutes it. (2) Complete the design. The current ablation has

bounded-branched, unbounded-branched and bounded-single arms but no unbounded-single arm at matched capacity, so branching and boundedness cannot be separated cleanly; a 2×2 grid at matched parameter count would. (3) Run the fingerprint-only and encoder-only arms, and report the block-influence statistic rather than accuracy alone. (4) Remove the scale-equivariance, by fixing ℓ or by moving to ARD, and ask whether the bound has any measurable effect once the median heuristic is not absorbing it. (5) Extend beyond binary assays. The kernel is agnostic to the label type, so regression endpoints and multi-label panels are the natural next benchmark, and they would test the geometry argument in a regime where the support set is not forced to be class-balanced.

O Feature Geometry and Molecular Attribution

Figure 13 gives in full the two views summarised in Section 5.7. A two-dimensional projection of a 2048-dimensional space is necessarily suggestive rather than conclusive, and we interpret it accordingly.

The upper row projects the mapped molecules from a Tox21 meta-test assay. Under BOND, the two classes begin to occupy distinguishable regions, with the circled clusters exhibiting visible boundaries. Under the baseline head, positive and negative molecules remain more intermingled.

The lower row examines the atom-level weights assigned to a single molecule. We regard the carbonyl and sulfonyl groups as informative substructures for this endpoint and bromine atoms and backbone carbons as uninformative. We report the counts because they provide more equivocal evidence than the projection. BOND with $M = 30$ recovers one of the two informative groups, whereas the baseline recovers neither. However, all three models place most of their weight on backbone carbons, and BOND with $M = 10$ recovers neither informative group. We therefore present this panel as qualitative support rather than quantitative evidence. It represents one molecule from one assay, and the comparison model was trained in an earlier experimental campaign rather than within the grid reported in the body.

Figure 14 extends the upper row of Figure 13 from one model to three, projecting the same Tox21 meta-test set under the baseline encoder, under the head with its bound removed, and under the full model. Under the full model the positives collect into a few tight groups at one edge of the cloud, while under the other two they stay dispersed among the negatives. We offer this as qualitative support for the claim that the head reshapes the space rather than as evidence that the reshaping is what produces the accuracy — Section 5.6 measures that the head does not equalise the two input modalities, and that the median-heuristic length-scale initialisation absorbs most of the scale variation the architecture was meant to control.

Figures 15 and 16 extend the lower row of Figure 13 from one molecule to a grid of them, drawn from the Tox21 meta-test split of 19,051 molecule-task pairs. Read across the sheets, the pattern the body figure asserts on a single molecule holds loosely rather than cleanly: weight concentrates on carbonyl, sulfonyl and other polar groups more often than chance, and is rarely placed on terminal halogens, but there are many molecules whose heaviest atoms are backbone carbons. We include the sheets because that mixed

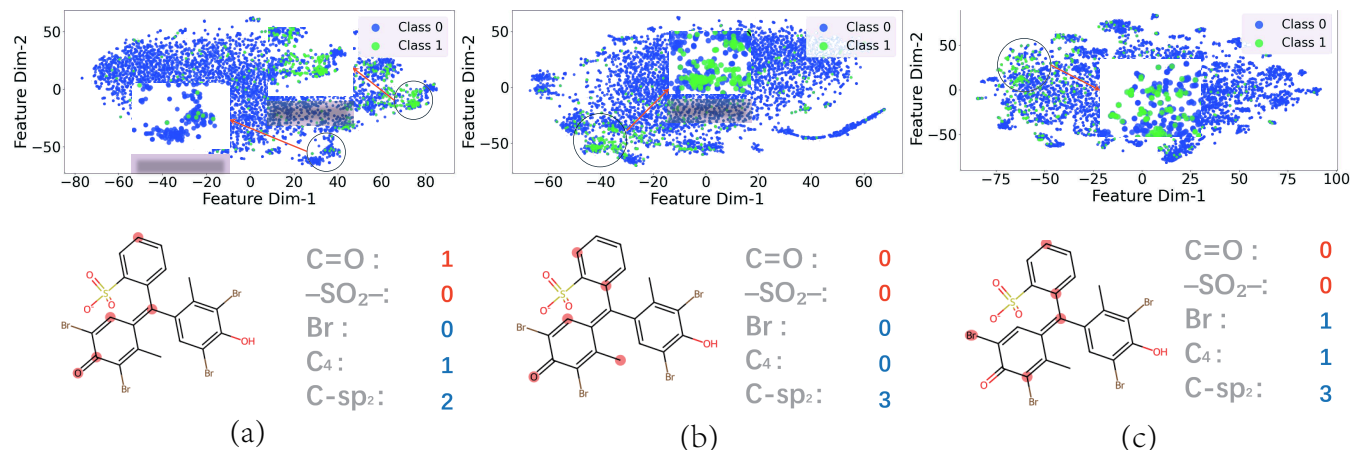


Figure 13: Feature-space projection (upper) and per-atom attribution (lower) for one Tox21 assay using (a) BOND with $M = 30$, (b) BOND with $M = 10$, and (c) the baseline head.

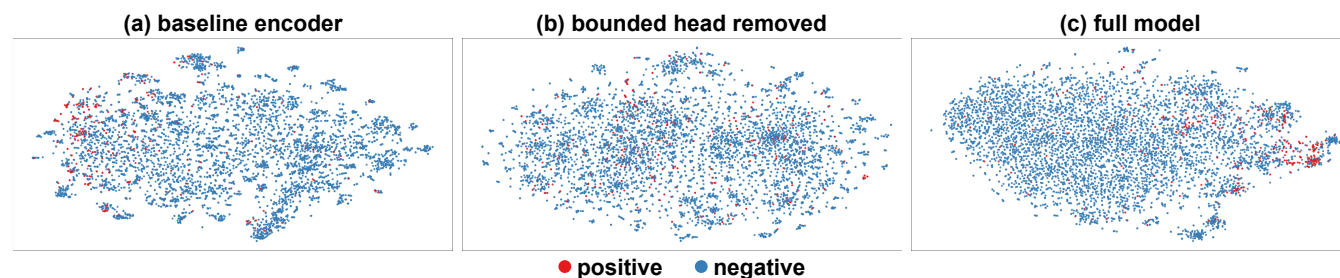


Figure 14: One Tox21 meta-test set in feature space under (a) the baseline encoder, (b) the head with its bound removed and (c) the full model. Positives concentrate under the full model and stay dispersed under the other two.

picture is what the model actually produces, and a reader assessing the single-molecule panel in the body should be able to see the population it was drawn from.

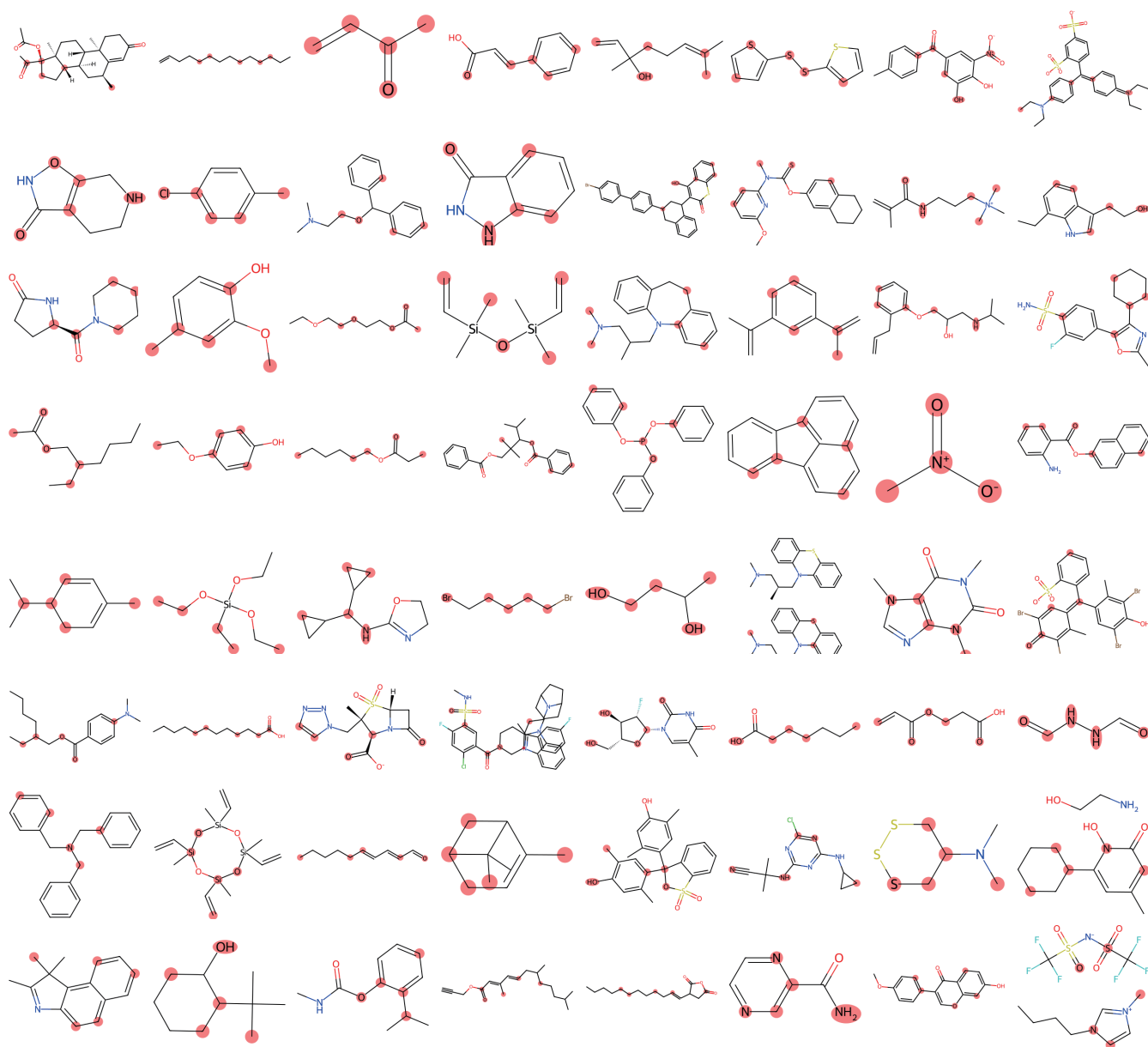
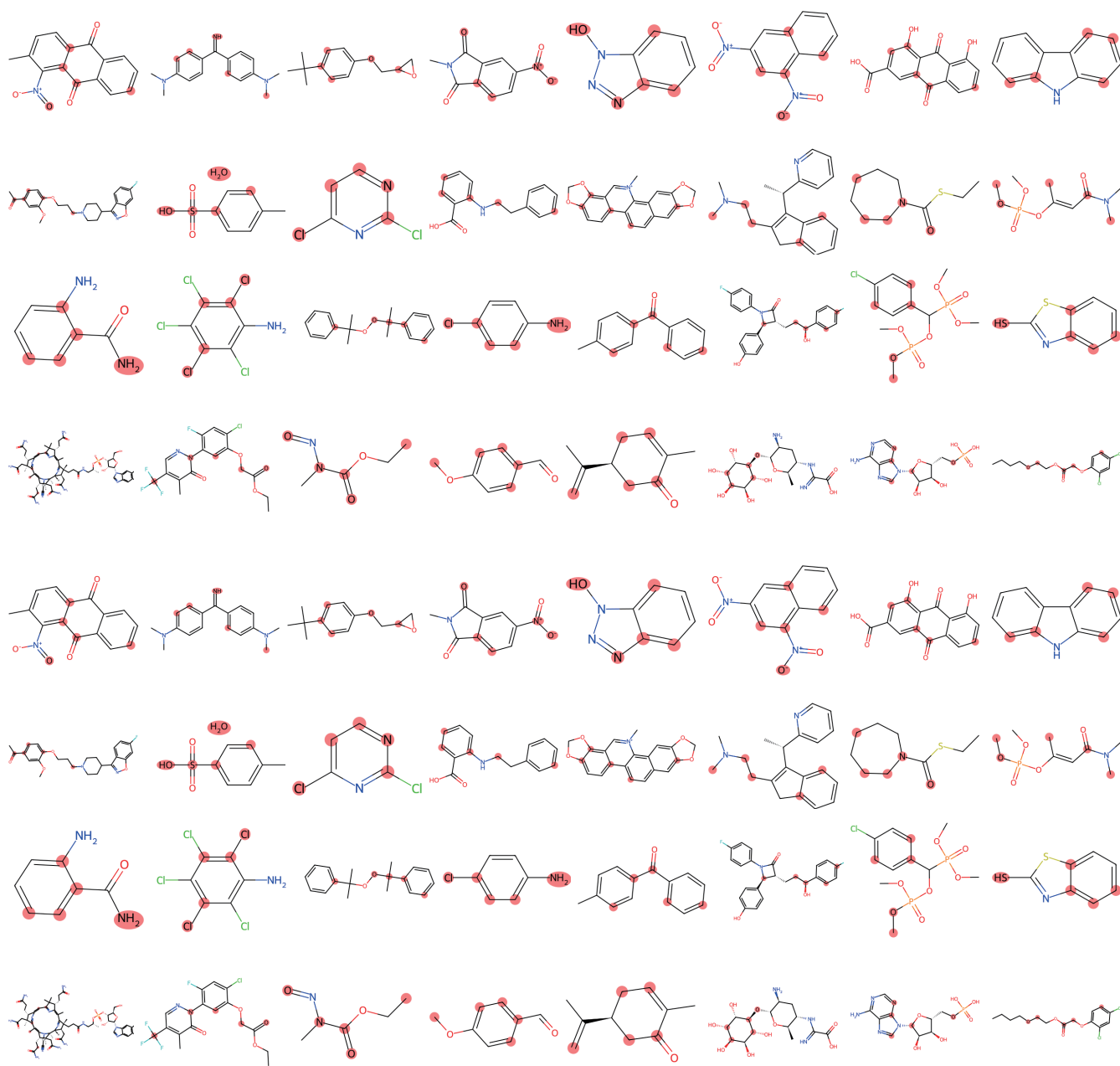


Figure 15: Per-atom attribution on Tox21 meta-test molecules, sheets 1 and 2 of 4.

**Figure 16: Per-atom attribution on Tox21 meta-test molecules, sheets 3 and 4 of 4.**

P References Cited in the Appendix

The works cited in this appendix are collected here so that it can be read on its own. They keep the order of the reference list after the main text and are numbered consecutively; citations in the appendix carry their number in that list.

- [1] Sami Abu-El-Haija, Bryan Perozzi, Amol Kapoor, Nazanin Alipourfard, Kristina Lerman, Hrayr Harutyunyan, Greg Ver Steeg, and Aram Galstyan. 2019. MixHop: Higher-order graph convolutional architectures via sparsified neighborhood mixing. In *International Conference on Machine Learning*. 21–29.
- [2] Thomas Adler, Johannes Brandstetter, Michael Widrich, Andreas Mayr, David Kreil, Michael Kopp, Günter Klambauer, and Sepp Hochreiter. 2020. Cross-domain few-shot learning by representation fusion. *arXiv preprint arXiv:2010.06498* (2020).
- [3] Han Altae-Tran, Bharath Ramsundar, Aneesh S Pappu, and Vijay Pande. 2017. Low data drug discovery with one-shot learning. *ACS Central Science* 3, 4 (2017), 283–293.
- [4] Wenlin Chen, Austin Tripp, and José Miguel Hernández-Lobato. 2023. Meta-learning adaptive deep kernel Gaussian processes for molecular property prediction. In *International Conference on Learning Representations*.
- [5] Chelsea Finn, Pieter Abbeel, and Sergey Levine. 2017. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning*. 1126–1135.
- [6] Zhichun Guo, Chuxu Zhang, Wenhao Yu, John Herr, Olaf Wiest, Meng Jiang, and Nitesh V. Chawla. 2021. Few-shot graph learning for molecular property prediction. In *Proceedings of the Web Conference*. 2559–2567.
- [7] Weihua Hu, Bowen Liu, Joseph Gomes, Marinka Zitnik, Percy Liang, Vijay Pande, and Jure Leskovec. 2020. Strategies for pre-training graph neural networks. In *International Conference on Learning Representations*.
- [8] Jongmin Kim, Taesup Kim, Sungwoong Kim, and Chang D Yoo. 2019. Edge-labeling graph neural network for few-shot learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11–20.
- [9] Gregory Koch, Richard Zemel, and Ruslan Salakhutdinov. 2015. Siamese neural networks for one-shot image recognition. In *ICML Deep Learning Workshop*, Vol. 2.
- [10] Ruifeng Li, Mingqian Li, Wei Liu, Yuhua Zhou, Xiangxin Zhou, Yuan Yao, Qiang Zhang, and Hongyang Chen. 2025. UniMatch: Universal matching from atom to task for few-shot drug discovery. In *International Conference on Learning Representations*.
- [11] Yanbin Liu, Juho Lee, Minseop Park, Saehoon Kim, Eunho Yang, Sung Ju Hwang, and Yi Yang. 2019. Learning to propagate labels: Transductive propagation network for few-shot learning. In *International Conference on Learning Representations*.
- [12] Johannes Schimunek, Philipp Seidl, Lukas Friedrich, Daniel Kuhn, Friedrich Rippmann, Sepp Hochreiter, and Günter Klambauer. 2023. Context-enriched molecule representations improve few-shot drug discovery. In *International Conference on Learning Representations*.
- [13] Jake Snell, Kevin Swersky, and Richard Zemel. 2017. Prototypical networks for few-shot learning. In *Advances in Neural Information Processing Systems*, Vol. 30.
- [14] Yaqing Wang, Abulikemu Abuduweili, Quanming Yao, and Dejing Dou. 2021. Property-aware relation networks for few-shot molecular property prediction. In *Advances in Neural Information Processing Systems*, Vol. 34. 17441–17454.
- [15] Zhenqin Wu, Bharath Ramsundar, Evan N Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S Pappu, Karl Leswing, and Vijay Pande. 2018.

MoleculeNet: a benchmark for molecular machine learning. *Chemical Science* 9, 2 (2018), 513–530.

- [16] Ruochi Zhang, Chao Wu, Qian Yang, Chang Liu, Yan Wang, Kewei Li, Lan Huang, and Fengfeng Zhou. 2024. MolFeSCue: enhancing molecular property prediction in data-limited and imbalanced contexts using few-shot and contrastive learning. *Bioinformatics* 40, 4 (2024), btae118.